Transportation Research Part C 179 (2025) 105286

Contents lists available at ScienceDirect

Transportation Research Part C

journal homepage: www.elsevier.com/locate/trc

Adaptive and flexible rail transit network service dispatching as a partially observable Markov decision process

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ARTICLE INFO

Keywords:
Transit routing
Service scheduling
Flexible fleet sizes
Markov decision process
Reinforcement learning

ABSTRACT

This paper presents a novel adaptive train scheduling framework with flexible fleet sizes for routing and scheduling in network-wide rail transit services. This framework aims to minimize both passenger waiting times and operating costs driven by prevailing passenger demand. The train scheduling problem is formulated as a partially observable Markov decision process (POMDP) to reflect the practicality in training and real-world applications. To address the computational challenges associated with the train scheduling problem, deep reinforcement learning techniques are applied to seek potential optimal solutions to the optimization problem. The proposed train scheduling framework is tested using real-world scenarios and the data collected from the Hong Kong Light Rail Transit (LRT) network. The experiment results demonstrate that the proposed train scheduling framework using flexible fleet sizes can effectively reduce passenger waiting time and operating costs. This study contributes to the real-time routing and scheduling of network-wide rail transit services by advanced optimization technology.

1. Introduction

Efficient rail transit systems are vital for alleviating traffic congestion and building sustainable urban cities. However, the heavy daily travel demand in large cities, particularly during peak hours, often leads to severe crowding at stations. The consequence significantly diminishes the attractiveness of urban rail transit systems (Rader et al., 2020). Meeting the high passenger demand and alleviating traffic congestion in busy rail transit networks can be regarded as a critical challenge for transit operators. In response to these challenges, many researchers have focused on demand-driven service scheduling problems with different routing strategies such as route choice, stop-skipping, and short-turning. The advantage of these strategies is their ability to improve service efficiency and passenger satisfaction without requiring additional train resources.

Several integer or mixed-integer programming models have been proposed for service routing and scheduling problems, as they can derive the global optimal solution from an academic perspective. For instance, Niu et al. (2015) proposed a mixed-integer programming model to minimize passenger waiting time through service routing and scheduling. Samà et al. (2017) presented a mixed-integer programming model for service routing and scheduling during disturbed railway traffic situations. Zhu and Goverde (2019) introduced a mixed-integer programming model for service rescheduling with flexible routing strategies during disruptions. Dong et al. (2020) developed a mixed-integer programming model to optimize the service scheduling and routing while considering passenger travel efficiency. Yuan et al. (2022) designed a mixed-integer programming model integrating service scheduling, routing, and rolling stock assignment for a metro line. Yuan et al. (2023) built a mixed-integer programming model to reduce passenger

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https://doi.org/10.1016/j.trc.2025.105286
Received 10 October 2024; Received in revised form 10 May 2025; Accepted 19 July 2025
Available online 5 August 2025

waiting time and station crowding through service routing and scheduling. Gong et al. (2024) proposed an integer programming model to minimize passenger waiting time and operating costs while considering rolling stock circulation. Tian et al. (2024) constructed an integer programming model for optimizing service routing, scheduling, and train type. Feng et al. (2024) developed a mixed-integer linear programming model to deal with the train service route design problem with passenger flow allocation. Chai et al. (2024a) proposed a mixed-integer linear programming model to optimize the routing and scheduling of train platoons. Although the effectiveness of integer or mixed-integer programming models has been demonstrated in various train routing and scheduling problems, solving integer programs may be computationally demanding and thus not readily applicable in real-time complex scenarios. Considering the complexity of practical applications, several studies on service routing and scheduling problems have investigated the use of more computationally efficient solution algorithms, such as genetic algorithms (GA, see Sun et al., 2014a, and Murali et al., 2016), variable neighborhood search (VNS, see Samà et al., 2017; Dong et al., 2020, and Wang et al., 2023b), and other variants (see Cacchiani et al., 2015; Jiang et al., 2017; Wang et al., 2021, and Tian et al., 2024). A comprehensive review of the application of meta-heuristic algorithms in service optimization can be found in Zhang et al. (2018).

Recently, various railway operators have implemented flexible fleet sizes to address the conflict between service capacity and dynamic passenger flow demand. Examples include Germany's ICE4, Beijing Capital Airport Express, Hong Kong MTR, and UK Network Rail (Yan et al., 2020; Ying et al., 2021). Some recent research has demonstrated that flexible fleet sizes can significantly improve the overall performance of services, including service schedule regularity and passenger waiting time (Ying et al., 2021, 2022; Pan et al., 2023; Chai et al., 2024a,b; Wang et al., 2024). Moreover, advancements in information and communication technologies, such as train positioning systems and smart card transaction systems, now enable the observation and reliable estimation of real-time variations in service status and passenger demand (Bešinović et al., 2021; Yin et al., 2022; Ying et al., 2024). These advancements present new opportunities and call for innovations in real-time demand-responsive train service scheduling with advanced data analytics and optimization techniques. However, incorporating flexible routing and a variable assigned fleet size of service runs significantly increases the complexity of the problem. When considering the routing and scheduling problem in network-wide rail transit with flexible fleet sizes, even the most advanced heuristic methods struggle to provide acceptable solutions in the short term or real-time. This issue is known as the curse of dimensionality, which is a major technical challenge faced in most dynamic optimization problems (Sutton and Barto, 2018; Bertsekas, 2019).

An emerging technique to address the above challenge is reinforcement learning (RL), which reduces the complexity of the original dynamic problem by approximating the decision and state spaces using parameter approximators like kernel regression models or artificial neural networks (Sutton and Barto, 2018; Wiering and Van Otterlo, 2012). Various RL-based approaches have been applied to optimize service operations. For example, Yin et al. (2016) and Liu et al. (2018) used approximate dynamic programming techniques to derive energy-efficient service rescheduling strategies. Ying et al. (2020) proposed an actor-critic deep reinforcement learning approach for real-time metro service scheduling under uncertain demand, which was later extended to incorporate flexible fleet sizes in Ying et al. (2022). Wang et al. (2023a) studied the application of multi-agent reinforcement learning methods in the real-time management of power supply shortages in subway systems. Yang et al. (2023) proposed a grid world model for a single-track railway scheduling problem and solved it by deep reinforcement learning.

Nevertheless, existing reinforcement learning approaches lack research on routing and scheduling problems in network-wide rail transit systems. The technical challenges include: (1) capturing the complex dynamics of trains and passengers in network-wide transit systems, (2) coordinating uneven passenger demand across different routes, (3) resolving potential conflicts in dispatching and track usage, and (4) managing flexible fleet sizes and rolling stock circulation within transit networks (Sun et al., 2014a; Zhou et al., 2021; Chai et al., 2024a).

To address the challenges mentioned above, this study proposes an adaptive train scheduling framework with flexible fleet sizes for routing and scheduling in network-wide rail transit services. The main contributions of this study can be summarized as follows:

- We develop a novel discrete event dynamic system model that accurately captures the dynamics of rail transit operations and passenger flows, even though some shared stations will be utilized by multiple routes. This model is a versatile framework applicable to transit networks involving different routing strategies, such as route choice, short-turning, and stop-skipping. Subsequently, the model is formulated as a partially observable Markov decision process (POMDP) (Powell, 2020) by introducing a novel and practical observation space to enhance its applicability in training and real-world scenarios. Additionally, we identify a series of operational constraints related to service dispatching and rolling stock circulation, particularly in situations where multiple routes depart from a single depot.
- We proposed an ANN-based surrogate function to approximate the future cost of the decisions made during the solution process. To incorporate the stage-dependent operational constraints on the decision variables, a novel masking scheme is devised associated with the surrogate function. Different training algorithms for the underlying ANN surrogate, including the deep Q network (DQN) and the deep recurrent Q network (DRQN) algorithms, were developed and tested to generate the real-time scheduling strategy that responds to prevailing demand and system variations. The computation results demonstrate the advantage of the DRQN algorithm in terms of result quality, validating the temporal dependencies of the train routing and scheduling problem considered in this study.
- The proposed train scheduling framework is tested using real-world scenarios and data collected from the Hong Kong Light Rail Transit (LRT) network. The comparative experiments demonstrate that the proposed framework outperforms the well-established meta-heuristic algorithm in terms of both computing times and the quality of the optimized solutions. Furthermore, additional experiments revealed that the proposed train scheduling framework using flexible fleet sizes can effectively reduce passenger waiting time and operating costs.

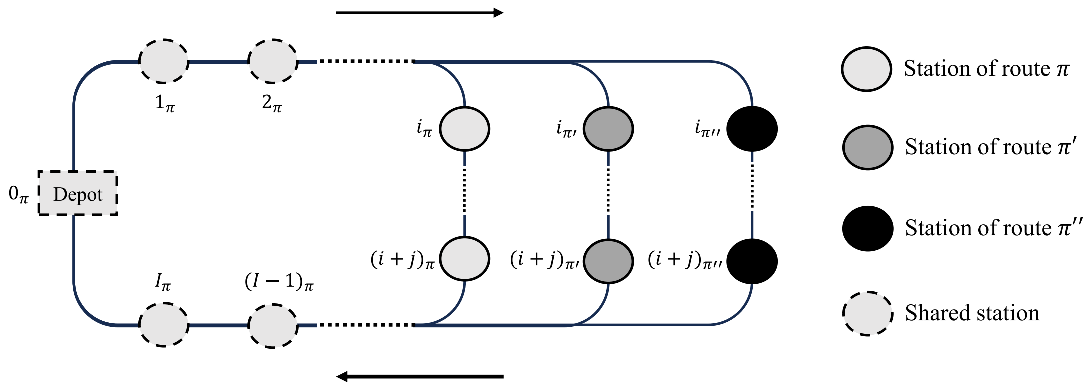


Fig. 1. A transit network example with three routes.

The structure of this paper is as follows: Section 2 introduces a discrete event dynamic system model of rail transit operation considering dynamic passenger flows. Section 3 presents the optimal train scheduling formulation as a partially observable Markov decision process with associated constraints. Section 4 describes the solution framework based on deep reinforcement learning algorithms. In Section 5, the solution framework is implemented and tested on a network-wide rail transit using data collected from the Hong Kong Light Rail Transit network. Finally, some concluding remarks and suggestions for future work are presented in Section 6.

2. Transit network model

In this section, we present a transit network model. The description of rail transit networks is provided in Section 2.1, followed by the objectives and assumptions adopted in this study, which are introduced in Section 2.2. Subsequently, the dynamics of service runs and the processes of passenger alighting and boarding within the system are discussed in Sections 2.3 and 2.4, respectively. For readers' reference, a list of notations employed in the transit network model can be found in Part 1 of Table 1.

2.1. Description of rail transit networks

We consider a rail transit network consisting of a set of routes denoted as Π. Each specific route $\pi \in \Pi$ consists of an ordered set of stations $\mathcal{I}_\pi = \{1_\pi, 2_\pi, \dots, i_\pi, \dots, (i+j)_\pi, \dots, (I-1)_\pi, I_\pi\}$. Here $\{i_\pi, \dots, (i+j)_\pi\}$ denotes the stations exclusive to a single route (i.e. non-shared stations), while $\{1_\pi, 2_\pi, \dots, (I-1)_\pi, I_\pi\}$ denotes the stations utilized by multiple routes (i.e. shared stations). Fig. 1 provides a basic example of a transit network with three routes. In this network, a service run on route π will depart from depot 0_π, follow the ordered set of stations $\mathcal{I}_\pi$ on route π towards the terminal station I_π, and then return to depot 0_π. It is noted that if all stations on route π are encompassed within the route π', then route π can be considered a short-turning or stop-skipping strategy of route π'. Furthermore, the total number of train units, including driving modules $\hat{G}$ and trailers $\hat{U}$, is limited, with the (re-)composition, deployment, and dispatching of service runs on all routes executed from a single depot.

To provide high-quality services to passengers in crowded transit networks, this study focuses on determining the dispatching headway and dispatching route for each service run. Instead of fixed fleet sizes, each service run departing from the depot consists of one driving module and a variable number of trailers. The advantage of this flexible fleet size strategy is that it enhances passenger satisfaction and helps mitigate congestion (Ying et al., 2022; Chai et al., 2024a). Consequently, the optimization problem involves determining the dispatching headway, dispatching route, and assigned fleet size for each service run, ensuring a real-time response to dynamic variations in passenger demand.

Passenger demand in the transit network is driven by a time series of origin–destination (OD) demand matrices over a defined time period T. Each element $O_{ij}(t)$ in the OD demand matrices represents the passenger arrival rate (i.e. arriving passengers per unit time) from an origin station i to a destination j during time interval $t \in [0, T]$ (Wong et al., 2008; Ying et al., 2020). Direct connections exist between stations i and j if they are on the same route; otherwise, passengers will have to make a transfer to reach their final destinations.

2.2. Objectives and assumptions

Rail transit operators aim to provide high-quality services to passengers while managing limited train resources and loading capacity, all while considering operating costs. During peak hours or in densely populated areas, passenger demand may exceed the loading capacity of service runs, leading to significant crowding at stations and causing inconvenience and dissatisfaction among passengers. To prevent passengers from being stranded at stations, flexible strategies are implemented in the rail system, including variable dispatching headways, dispatching routes, and assigned fleet sizes. However, these strategies inevitably impact the operating

Table 1
Notations used in the transit network model and optimal train scheduling system.

Notation	Definition
Part 1: Notations used in the transit network model.	
Indices	
t	Index of time interval.
n	Index of decision stage (and service run).
m	Index of service run departing from the depot.
M, K	Index of train service passing through a specific station.
π, π', π''	Indices of routes.
i, j, k	Indices of stations in the transit network.
i_π, j_π	Indices of stations on route π.
Sets	
Π	Set of routes.
$\mathcal{I}$	Set of stations.
$\mathcal{I}_\pi$	Set of stations on route π.
$\boldsymbol{\sigma}_{n,i}$	Set of departure times for station i, considering n service runs departing from the depot.
$\hat{\boldsymbol{\sigma}}_{n,i}$	Set of departure times for station i sorted in ascending order, considering n service runs departing from the depot.
$\mathcal{H}^n_{M,i_\pi}$	Set of service runs on other routes from which alighted passengers could board the Mth train service at station i_π.
Parameters	
T	Length of service period.
Π	Total number of routes.
$\hat{G}$	Initial number of driving modules in the depot 0.
$\hat{U}$	Initial number of trailers in the depot 0.
$\hat{C}$	Carrying capacity of a driving module or a trailer.
Environment variables	
Λ_n	Carrying capacity of service run n.
τ_{n,i_π}	Arrival time of service run n at station i_π on route π.
σ_{n,i_π}	Departure time of service run n from station i_π on route π.
$\hat{\sigma}^n_{K,i_\pi}$	Last departure time at station i_π in the departure time set $\hat{\boldsymbol{\sigma}}_{n,i}$.
z_{i_π}	Running time between stations $(i-1)_\pi$ and i_π on route π.
w_{i_π}	Dwell time at station i_π on route π.
$\Delta_{i'_{\pi'},i_\pi}$	Transfer time from station $i'_{\pi'}$ to station i_π.
$O_{ij}(t)$	Demand rate from station i to station j over time interval t.
$\hat{O}^k_{i_\pi j}(t)$	A realization of OD demand from station i_π to station j, while having to alight at station k for transfer.
$\Omega_{i_\pi,j}(\bar{n})$	Number of passengers alighting from service run $\bar{n}$, aiming to transfer to station i_π for station j.
$\lambda^n_{M,i_\pi j}$	Number of waiting passengers to board Mth train service at station i_π for station j, considering n service runs departing from the depot.
$\hat{\lambda}^n_{M,i_\pi j}$	Actual number of passengers who can board the Mth train service at station i_π for station j, considering n service runs departing from the depot.
$l^n_{M,i_\pi j}$	Number of remaining passengers at station i_π traveling to station j following the departure of Mth train service, considering n service runs departing from the depot.
l^n_{M,i_π}	Total number of remaining passengers at station i_π following the departure of Mth train service, considering n service runs departing from the depot.
ρ^n_{M,i_π}	Total number of on-board passengers when Mth train service departs from its previous station, considering n service runs departing from the depot.
v^n_{M,i_π}	Total number of passengers alighting from Mth train service at station i_π, considering n service runs departing from the depot.
$b^n_{M,i_\pi j}$	Proportion of $\lambda^n_{M,i_\pi j}$ out of all passengers who want to board the Mth train service at station i_π for station j, considering n service runs departing from the depot.
Part 2: Notations used in optimal train scheduling system.	
Sets	
$\boldsymbol{s}_n$	State set at stage n.
$\boldsymbol{o}_n$	Observation set at stage n.
$\boldsymbol{x}_n$	Decision set at stage n.

(*continued on next page*)

cost. Specifically, more frequent dispatching and larger assigned fleet sizes can reduce passenger waiting times, but will increase the cost for transit operators. Therefore, this study aims to minimize both the total passenger waiting time and the operating cost of the rail transit network.

This study also adopts the following assumptions regarding the transit network setting:

Table 1 (*continued*).

Notation	Definition
Parameters	
α	Trade-off coefficient between perspectives of passengers' and operators'.
γ	Discount factor used in the optimization formulation.
ζ	Coefficient to adjust the operational cost of trains.
Δ_S	Safety margin between any service run pair.
Δ_T	Slack time required for redeploying the returned train.
ζ^W	Unit monetary cost of passenger waiting time.
ζ^F	Unit fuel cost.
ζ^D	Unit non-fuel cost.
M^{train}	Tare mass of a train unit.
M^{psg}	Average mass of a single passenger.
$\beta_1, \beta_2, \beta_2$	Coefficients of the Davis formula.
u_{max}	Maximum fleet sizes for each train service.
h_{min}, h_{max}	Lower and upper bounds of the dispatching headways.
H_{min}, H_{max}	Each route's lower and upper bounds of the dispatching headways.
Decision variables	
π_n	Dispatching route of service run n.
u_n	Fleet sizes of service run n.
h_n	Headway between service run $n-1$ and n.
Environment variables	
η_n	Terminal indicator at stage n.
r_n	Reward signal at stage n.
r_n^O	Reward signal for the operator at stage n.
r_n^W	Reward signal for passengers at stage n.
v_{i_π}	Speed between stations i_π and $(i+1)_\pi$.
$d_{i,i+1}^{\pi(M)}$	Distance between stations i_π and $(i+1)_\pi$.
T_π	Total journey time of route π.
$\hat{U}_t^n$	Number of the available fleet at time t associated with service run n.
$\beta_i(t)$	Number of waiting passengers at station i at time t.
$W_{M,i}^n$	Passengers waiting times at station i from $(M-1)$th to Mth train service at stage n.
$W_{total,i}^n$	Cumulative passenger waiting cost at station i at stage n.
W_{total}^n	Cumulative passenger waiting cost at stage n.
$O_{total,i}^n$	Cumulative operating cost at station i at stage n.
O_{total}^n	Cumulative operating cost at stage n.
Functions	
$\boldsymbol{P}$	Transition function from stage n to stage $n+1$.
$\boldsymbol{R}$	Reward function from stage n to stage $n+1$.

1. For safety reasons, each station can only accommodate one train service at a time, thereby preventing any collisions between train services during operations (Ying et al., 2020; Nguyen and Chow, 2023).
2. As a typical characteristic of rail transit operations, train running times through the network are assumed to be free from congestion and disturbances (Guo et al., 2017; Ying et al., 2022).
3. Passengers tend to board the first train service with available capacity to take them to their intended destinations (Chow et al., 2024).
4. Passengers are expected to transfer at stations with the minimum number of transfers and journey time (Chow et al., 2019; Nguyen et al., 2021).
5. Passengers cannot board a train service that has reached its maximum carrying capacity. In such cases, waiting passengers queue up at the station and board the next available train service following the multi-commodity first-in-first-out (FIFO) principle (Chow, 2009; Ying et al., 2024).

Assumption 1 aligns with the requirements of most rail transit systems due to their simple platform layout (Ying et al., 2020; Yuan et al., 2022). Assumption 2 is widely adopted in train service scheduling tasks, as advanced automatic train scheduling systems can quickly recover the nominal timetable from minor delays caused by factors such as train door problems or weather-induced speed restrictions (Sun et al., 2014b; Shi et al., 2018; Yuan et al., 2022). Assumption 3 is a reasonable assumption to simplify the problem, considering the high departure frequency of urban transit systems and the urgent travel needs of passengers during peak hours (Gu et al., 2016; Fan and Ran, 2021). Assumption 4 has been verified by travel behavior surveys conducted by MTR and adopted by many other researchers (Nikolić and Teodorović, 2014; Canca et al., 2019; Ying et al., 2022). Furthermore, Assumption 5 is a reasonable simplification of passenger boarding behavior.

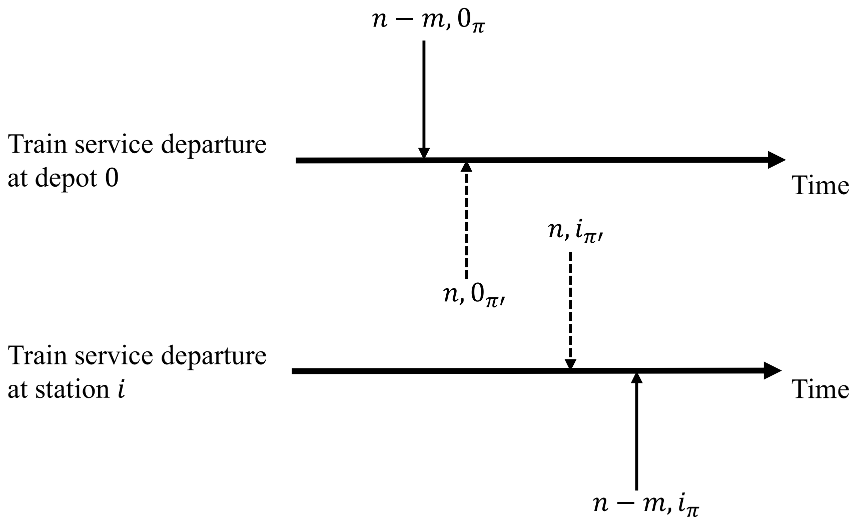


Fig. 2. Illustration of the sequence of train services passing through the shared station i.

2.3. Dynamic of service runs

We now introduce the dynamics of service runs in the transit network. Given the fleet sizes u_n of service run n, we can first have the carrying capacity Λ_n of the train service as follows:

$$\Lambda_n = \hat{C} \times u_n, \tag{1}$$

where the parameter $\hat{C}$ represents the carrying capacity of each train unit, which can be either a driving module or a trailer.

Given the dispatching headway h_n between service run $n-1$ and n, the dispatching time $\sigma_{n,0}$ of service run n from the shared depot 0 can be determined as:

$$\sigma_{n,0} = \sigma_{n-1,0} + h_n. \tag{2}$$

For service run n with the chosen dispatching route π, the corresponding arrival time τ_{n,i_π} and departure time σ_{n,i_π} at station i_π based on Assumption 2 as follows:

$$\tau_{n,i_\pi} = \sigma_{n,(i-1)_\pi} + z_{i_\pi}, \tag{3}$$

and

$$\sigma_{n,i_\pi} = \tau_{n,i_\pi} + w_{i_\pi}, \tag{4}$$

where z_{i_π} represents the nominal running time between adjacent station pairs $((i-1)_\pi, i_\pi)$, and w_{i_π} represents the nominal dwell time for station i_π.

Furthermore, in the considered transit network, the sequence in which train services pass through shared stations may differ from the sequence in which they depart from the depot. This situation is illustrated in Fig. 2. In the figure, train service $n-m$ departs from the depot 0 and follows route π, while train service n also departs from the same depot but follows route π'. Due to the different routes these two train services take, their journey times to reach the shared station i vary. As a result, even though train service $n-m$ departs from the depot 0 earlier than train service n, it arrives at station i after train service n.

To coordinate the sequence of service runs at shared stations, we first define the departure time set $\boldsymbol{\sigma}_{n,i}$ for a specific station $i \in \mathcal{I}$ as:

$$\boldsymbol{\sigma}_{n,i} = \begin{cases} \boldsymbol{\sigma}_{n-1,i} \cup \{\sigma_{n,i_\pi}\}, & \text{if } \exists\ i_\pi = i, \\ \boldsymbol{\sigma}_{n-1,i}, & \text{otherwise}, \end{cases} \tag{5}$$

where the departure time set $\boldsymbol{\sigma}_{n,i}$ will incorporate the departure time σ_{n,i_π} of service run n if there exists a station i_π for route π that corresponds to the same geographic location as station i. Furthermore, the elements of the set $\boldsymbol{\sigma}_{n,i}$ can be sorted in ascending order, resulting in the sorted departure time set:

$$\hat{\boldsymbol{\sigma}}_{n,i} = \{\hat{\sigma}^n_{1,i_\pi}, \hat{\sigma}^n_{2,i_{\pi'}}, \dots, \hat{\sigma}^n_{K,i_{\pi''}}\}, \tag{6}$$

where K represents the total number of departure times in the set $\hat{\boldsymbol{\sigma}}_{n,i}$. It is important to note that the stations $(i_\pi, i_{\pi'}, \dots, i_{\pi''})$ corresponds to the same geographic location as station i. The variables $\hat{\sigma}^n_{1,i_\pi}$ and $\hat{\sigma}^n_{K,i_{\pi''}}$ in (6) respectively represent the earliest and the latest departure times at station i in the departure time set $\hat{\boldsymbol{\sigma}}_{n,i}$.

The formulas (1)–(6) specify the dynamics of service runs considered in this study.

2.4. Evolution of passenger flow

In addition to the evolution of service runs, we also need to consider the processes of passenger boarding and alighting at stations. Assuming that service run n is the Mth train service passing through station i_π, the number of waiting passengers to board Mth train service $(1 \leq M \leq K)$ at station i_π for their destination j can be expressed as:

$$\lambda^n_{M,i_\pi j} = \begin{cases} \int_{\hat{\sigma}^n_{M-1,i_{\pi'}}}^{\hat{\sigma}^n_{M,i_\pi}} \hat{O}^k_{i_\pi j}(t)dt + l^n_{M-1,i_\pi j} + \sum_{\forall \bar{n} \in \mathcal{H}^n_{M,i_\pi}} \Omega_{i_\pi,j}(\bar{n}), & \text{if } j \in \pi \cup k \in \pi, \\ 0, & \text{otherwise,} \end{cases} \tag{7}$$

in which $\hat{O}^k_{i_\pi j}(t)$ is a specific realization of OD demand from station i_π on route π to station j in the transit network, while having to alight at station k for transfer. It is noted that stations j and k could refer to the same station if there is a direct connection between stations i_π and j via route π. The second term $l^n_{M-1,i_\pi j}$ refers to the number of passengers remaining at station i_π after the departure of $(M-1)$th train service, who still intends to travel to station j.

Moreover, we explicitly incorporated the transfer passenger demand through the last summation term $\sum_{\forall \bar{n} \in \mathcal{H}^n_{M,i_\pi}} \Omega_{i_\pi,j}(\bar{n})$ in (7). Specifically, we define a condition set $\mathcal{H}^n_{M,i_\pi}$ (see also Nguyen et al., 2021; Ying et al., 2022). A service $\bar{n}$ is included in this set if:

$$\hat{\sigma}^n_{M-1,i_{\pi'}} < \tau_{\bar{n},k''_{\pi''}} + \Delta_{k''_{\pi''},i_\pi} \leq \hat{\sigma}^n_{M,i_\pi} \tag{8}$$

where the notation $\tau_{\bar{n},k''_{\pi''}}$ represents the arrival time of service run $\bar{n}$ at station $k''_{\pi''}$, and the notation $\Delta_{k''_{\pi''},i_\pi}$ represents the transfer time from station $k''_{\pi''}$ to station i_π. The set $\mathcal{H}^n_{M,i_\pi}$ hence includes all service runs that satisfy the transfer requirements described in (8). In addition, the notation $\Omega_{i_\pi,j}(\bar{n})$ denotes the number of passengers alighting from service run $\bar{n}$ at station $k''_{\pi''}$ on some other route π'', aiming to transfer to station i_π for their destination j. Consequently, the summation term $\sum_{\forall \bar{n} \in \mathcal{H}^n_{M,i_\pi}} \Omega_{i_\pi,j}(\bar{n})$ gives the total number of passengers arriving at station i_π from other routes, intending to board service run M for their destination j.

Furthermore, the actual number of passengers who can board the Mth train service at station i_π for station j is determined as:

$$\hat{\lambda}^n_{M,i_\pi j} = min\{\lambda^n_{M,i_\pi j}, b^n_{M,i_\pi j}[\Lambda_M - \rho^n_{M,(i-1)_\pi} + v^n_{M,i_\pi}]\}, \tag{9}$$

where

$$b^n_{M,i_\pi j} = \frac{\lambda^n_{M,i_\pi j}}{\sum_{\forall j \in \pi \cup k \in \pi} \lambda^n_{M,i_\pi j}} \tag{10}$$

is the proportion of $\lambda^n_{M,i_\pi j}$ out of all passengers who want to board the Mth train service at station i_π for station j or station k.

The term Λ_M in (9) is the carrying capacity of Mth train service (refer to (1)). Furthermore, the notation $\rho^n_{M,(i-1)_\pi}$ represents the number of on-board passengers when Mth train service departs from its previous station $(i-1)_\pi$. It can be updated as:

$$\rho^n_{M,i_\pi} = \rho^n_{M,(i-1)_\pi} - v^n_{M,i_\pi} + \hat{\lambda}^n_{M,i_\pi}, \tag{11}$$

where $\hat{\lambda}^n_{M,i_\pi} = \sum_{\forall j \in \pi \cup k \in \pi} \hat{\lambda}^n_{M,i_\pi j}$ is the total number of passengers who can board the Mth train service at station i_π. Furthermore, the notation v^n_{M,i_π} denotes the number of passengers alighting from Mth train service at station i_π. It can be determined as:

$$v^n_{M,i_\pi} = \sum_{\forall i' \in \pi} \hat{\lambda}^n_{M,i' i_\pi}, \tag{12}$$

in which i' represents all preceding stations heading to station i_π in route π.

The last term $[\Lambda_M - \rho^n_{M,(i-1)_\pi} + v^n_{M,i_\pi}]$ in (9) hence represents the remaining capacity on Mth train service to accommodate boarding passengers at station i_π. The formulations presented in (9) and (10) essentially allocates available space on the transit vehicle based on the demand split specified in the origin–destination (OD) matrices. This aligns with the multi-commodity FIFO principle in Assumption 5 (see Chow, 2009; Ying et al., 2020, 2024) which is often used in macroscopic network modeling. The principle implies the demand flows between each origin and destination pair would be treated separately and equally when the capacity is limited (i.e. they do not have priority over each other) and the serving order of the passengers is simply based upon the arrival order of the passengers regardless of which origin–destination pair they are associated with. It should be noted that the assumption of this (multi-commodity) FIFO would not affect the applicability of the proposed optimization framework while one would need to track the movement of each individual passenger through the network in a microscopic sense should this FIFO assumption was not made. It is practically achievable with construction of more sophisticated models, but that would give rise to extra computational burden and complexity in the modeling and performance evaluation process.

We can then determine the number of remaining passengers after Mth train service departs from station i_π as follows:

$$l^n_{M,i_\pi} = \sum_{\forall j \in \pi \cup k \in \pi} (\lambda^n_{M,i_\pi j} - \hat{\lambda}^n_{M,i_\pi j}) + \sum_{\forall j \notin \pi \cap k \notin \pi} \left(\int_{\hat{\sigma}^n_{M-1,i_{\pi'}}}^{\hat{\sigma}^n_{M,i_\pi}} \hat{O}^k_{i_\pi j}(t)dt + \sum_{\forall \bar{n} \in \mathcal{H}^n_{M,i_\pi}} \Omega_{i_\pi,j}(\bar{n}) \right). \tag{13}$$

The last sum formula in (13) represents the total number of passengers who arrive between the time $\hat{\sigma}^n_{M-1,i_{\pi'}}$ and $\hat{\sigma}^n_{M,i_\pi}$ but have to wait at station i_π because the Mth train service is unable to take them to their destinations.

The formulas (7)–(13) specify the evolution of passenger flow considered in this study.

3. Optimal train scheduling system

In this section, we present the formulation of the optimal train scheduling system. The notation used in this formulation is summarized in Part 2 of Table 1. The following subsections first introduce the formulation of the Markov decision process (MDP) and partially observable Markov decision process (POMDP) for the train scheduling system, followed by the definition of the reward signal, and finally, the state-dependent operational constraints imposed on the scheduling system.

3.1. MDP and POMDP formulation

At each stage n, the optimal train scheduling system is characterized by a tuple of state variables $\boldsymbol{s}_n$ and influenced by a tuple of decision variables $\boldsymbol{x}_n$. In this study, the state set is defined as $\boldsymbol{s}_n = (l^{n-1}_{0,i_\pi}, \dots, l^{n-1}_{K,i_\pi}, u_0, \dots, u_{n-1}, \hat{\boldsymbol{\sigma}}_{n-1,i})$, which includes the number of remaining passengers $(l^{n-1}_{0,i_\pi}, \dots, l^{n-1}_{K,i_\pi})$ at each station after the last train service has passed, the assigned fleet sizes of previous service runs $(u_0, \dots, u_{n-1})$, and the departure time set $\hat{\boldsymbol{\sigma}}_{n-1,i}$ at each station at stage $n-1$. Associate with the state, the decision vector is defined as $\boldsymbol{x}_n = (\pi_n, u_n, h_n)$, which consists of the chosen dispatching route π_n of service run n, the assigned fleet sizes u_n of service run n, and the headway between service runs $n-1$ and n.

At a given decision stage n, the Markovian system is in a specific state $\boldsymbol{s}_n$, and service run n is dispatched based on the decision $\boldsymbol{x}_n$. Following the execution of this decision, the system transitions to the next state $\boldsymbol{s}_{n+1}$ according to the transition function $\boldsymbol{s}_{n+1} \sim \boldsymbol{P}(\boldsymbol{s}_n, \boldsymbol{x}_n)$, and receives a reward r_n from the reward function $r_n \sim \boldsymbol{R}(\boldsymbol{s}_n, \boldsymbol{x}_n)$. Specifically, the execution of decision $\boldsymbol{x}_n$ in state $\boldsymbol{s}_n$ involves several system transitions: the assigned fleet size u_n and the carrying capacity Λ_n of service run n is recorded based on (1); departure times σ_{n,i_π} and arrival times τ_{n,i_π} of service run n at each station are calculated using (2)–(4); and the sorted departure time set $\hat{\boldsymbol{\sigma}}_{n,i}$ is updated according to (5)–(6). Subsequently, the number of remaining passengers at each station is updated, reflecting passenger flow dynamics after the completion of service run n, as described by (7)–(13). These dynamics, established through (1)–(13), complete the specifications of the transition from state $\boldsymbol{s}_n$ to $\boldsymbol{s}_{n+1}$. Finally, the system terminates if the binary variable $\eta_n = 1$, which indicates all passenger demand in the transit network has been served at stage n; otherwise, $\eta_n = 0$.

From the definition of state variables, it can be observed that the dimensions of the state space increase with each stage n. This is due to the fact that the sequence in which train services pass through shared stations may differ from the sequence in which they depart from the depot (see Fig. 2). To accurately capture passenger boarding and alighting details, the state must include not only the dispatching information of the current service run but also that of previous service runs, as well as historical passenger-related information. As a result, the state space becomes both large and dynamic, presenting a significant challenge when applying reinforcement learning in real-world scenarios (Liu et al., 2015; Dulac-Arnold et al., 2021). While we address the state space challenge through advanced feature extraction techniques and parametric approximators in a simulation environment, the occasional unavailability of historical passenger-related information in real-world applications can negatively affect the model's performance.

In such cases, a partially observable Markov decision process (POMDP) (Powell, 2020) is considered in our study. In a POMDP, the states $\boldsymbol{s}_n$, decisions $\boldsymbol{x}_n$, transition function P, and reward function R remain the same as in a standard MDP. The key difference is that the controller no longer has direct access to the full system state. Instead, the controller receives an observation generated from the underlying system state and makes decisions based on this observation. In this study, we define the observation set $\boldsymbol{o}_n = (l^{n-1}_{K,i_\pi}, \hat{\sigma}^{n-1}_{K,i_\pi})$, in which the notation l^{n-1}_{K,i_π} represents the number of remaining passengers at each station after the last train service has passed, and $\hat{\sigma}^{n-1}_{K,i_\pi}$ represents the last departure time at each station at stage $n-1$. By transforming the MDP into a POMDP, the controller receives a smaller, fixed observation set, which reduces the complexity of the state space. This transformation makes the controller more suitable for real-world applications.

3.2. Reward signal

We now introduce the reward signal to reflect the objective for the Markovian system, which consists of two components: one related to passengers and the other related to operators (Li and Lo, 2014; Chow et al., 2017).

From the passengers' perspective, the objective is to minimize the cost associated with passenger waiting time at stations. The process of a passenger boarding and queueing at a specific station i can be illustrated by Fig. 3 (see also Chow and Pavlides, 2018; Daganzo and Ouyang, 2019). In the figure, the solid line represents the cumulative number of passengers arriving at station i over time, while the dashed line represents the cumulative number of passengers served over time. The vertical difference $\beta_i(t)$ represents the number of waiting passengers at station i over time. With the assumption of FIFO made in Assumption 5, the total waiting time of passengers from sorted departure time $\hat{\sigma}^n_{M-1,i_{\pi'}}$ to $\hat{\sigma}^n_{M,i_\pi}$ can be calculated as the area between the cumulative arrivals and departures (Chow and Pavlides, 2018; Daganzo and Ouyang, 2019):

$$W^n_{M,i} = \int_{\hat{\sigma}^n_{M-1,i_{\pi'}}}^{\hat{\sigma}^n_{M,i_\pi}} \beta_i(t)dt. \tag{14}$$

It should be noted that the assumption of FIFO would not affect the applicability of the scheduling system proposed herein. Nevertheless, one would need to track and add up waiting times of each individual passenger in a microscopic sense should FIFO assumption was not made, which would give rise to extra computational burden and complexity in modeling and the performance evaluation process.

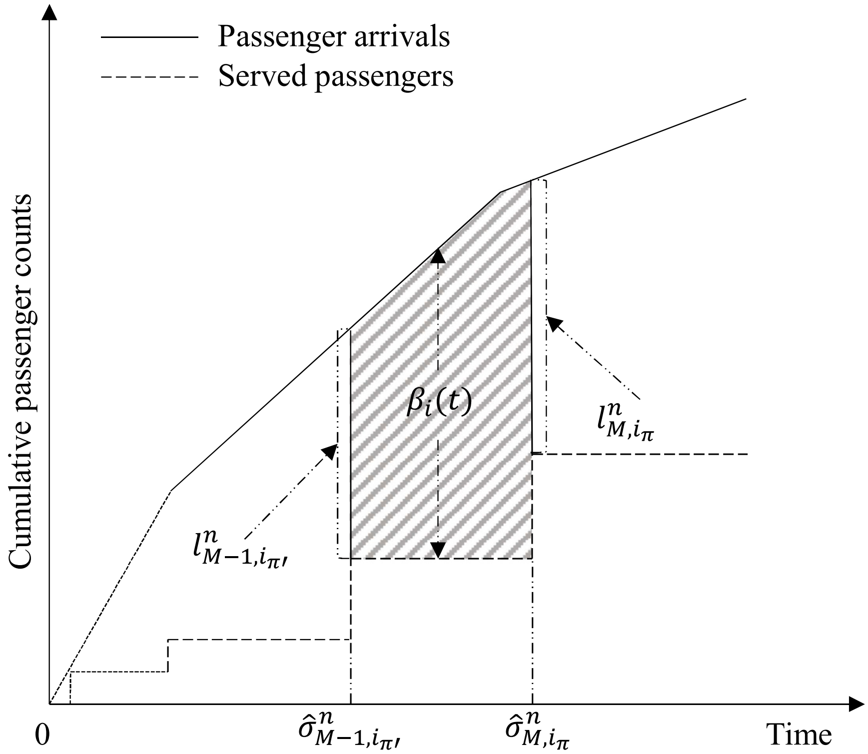


Fig. 3. Boarding and queueing process at a specific station.

Furthermore, the cumulative passenger waiting cost W^n_{total} in the transit network at stage n can be calculated as follows:

$$W^n_{total} = \sum_{\forall i \in \mathcal{I}} W^n_{total,i}, \tag{15}$$

where

$$W^n_{total,i} = \sum_{K_0=1}^{K_1} \zeta^W W^n_{K_0,i} \tag{16}$$

is the cumulative passenger waiting cost at station i from the first train service to the K_1th train service, and ζ^W is the coefficient representing an appropriate monetary cost per unit of time (Chow and Pavlides, 2018). It is noted that K_1 denotes the total number of service runs that pass through station i after service run n completes its service.

The reward signal for passengers r^W_n at stage n is then defined as:

$$r^W_n = W^n_{total} - W^{n-1}_{total}. \tag{17}$$

This reward signal ensures that the total passenger waiting cost obtained after system termination reflects the actual reality by subtracting a cost that may not align with the actual passenger waiting cost.

From the operators' perspective, we first consider the energy cost of service runs, which is influenced by the traction required by the train service to overcome resistance during its progress. Given the speed $v_{i_\pi} = \frac{d^\pi_{i,i+1}}{z_{i_\pi}}$ between stations i_π and $(i+1)_\pi$, the traction of Mth train service can be calculated by the Davis formula (Davis, 1926; Hansen et al., 2017):

$$F(M, v_{i_\pi}) = (\beta_1 + \beta_2 v_{i_\pi} + \beta_3 v^2_{i_\pi})(M^{train} u(M) + M^{psg} \rho^n_{M,i_\pi}), \tag{18}$$

in which M^{train} represents the tare mass of a train unit, M^{psg} represents the average mass of a single passenger, and $u(M)$ represents the assigned fleet sizes for the Mth train service. If the assigned fleet sizes for the service run increase, the tare mass and maximum capacity of the service run will also increase, leading to higher energy costs and operating costs. Additionally, the coefficients β_1, β_2, and β_3 are empirical parameters that depend on the specific characteristics of the train being used.

Furthermore, the cumulative operating cost O^n_{total} in the transit network at stage n can be calculated as follows:

$$O^n_{total} = \sum_{\forall i \in \mathcal{I}} O^n_{total,i}, \tag{19}$$

where

$$O^n_{total,i} = \sum_{K_0=1}^{K_1} \zeta \left(\zeta^F F(K_0, v_{i_\pi}) + \zeta^D\right) d^\pi_{i,(i+1)} \tag{20}$$

is the cumulative operating cost at station i from the first train service to the K_1th train service, and ζ^F and ζ^D are the unit price of the fuel and the unit non-fuel cost related to distance, respectively. Furthermore, the coefficient ζ is introduced to adjust the operating cost of service runs, thereby accounting for the impact of factors such as track gradient, curve, and other related elements.

The reward signal for the operator r^O_n at stage n is then defined as:

$$r^O_n = O^n_{total} - O^{n-1}_{total}. \tag{21}$$

This reward signal ensures that the total operating cost obtained after system termination reflects the actual reality by subtracting a cost that may not align with the actual operating cost.

Finally, the reward signal r_n can be formulated as:

$$r_n = -\alpha r^W_n - (1-\alpha) r^O_n, \tag{22}$$

where α is the coefficient that adjusts the trade-off between the rewards of passengers and operators.

With the defined reward signal, the objective of the routing and scheduling problem is to maximize the sum of the future discounted reward from the current step n_0 over a potentially infinite horizon n^T (Bertsekas, 2019; Chow et al., 2021):

$$\max_{\boldsymbol{x}_n} \boldsymbol{R} = \lim_{n^T \to \infty} \sum_{n=n_0}^{n_0+n^T} \gamma^n r_n \tag{23}$$

via seeking the appropriate decisions $\boldsymbol{x}_n$ for all service runs. Here, $\gamma \in (0,1)$ represents the discount factor, which indicates the relative preference for immediate rewards compared to future rewards.

3.3. Operational constraints

The decision variables $\boldsymbol{x}_n$ in (23) are subject to operational and regulatory constraints. We first impose boundary constraints on fleet sizes and dispatching headways as follows:

$$1 \le u_n \le u_{max}, \tag{24}$$

$$h_{min} \le h_n \le h_{max}, \tag{25}$$

$$u_n, h_n \in \mathbb{Z}^+, \tag{26}$$

in which u_{max} indicates the maximum fleet sizes for each service run, while h_{min} and h_{max} represent the lower and upper bounds on the dispatching headways, respectively. The notation $\mathbb{Z}^+$ represents the set of positive integers.

Furthermore, assuming that the chosen dispatching route for service run n is π, it is necessary to satisfy the following boundary constraints to maintain the level of service on a specific route $\pi \in \Pi$:

$$H_{min} \le \sigma_{n,0_\pi} - \sigma_{n-k,0_\pi} \le H_{max}, \tag{27}$$

where $n-k$ represents the last service run that departs on route π among the past dispatches. The notations H_{min} and H_{max} represent each route's lower and upper bounds of the dispatching headways.

To resolve the potential conflict between constraints (25)–(27), we establish the following propositions. Proposition 3.1 derives the necessary condition for a feasible solution to the train scheduling problem. Proposition 3.2 provides a regulatory upper bound to ensure the feasibility of the dispatching time of service runs.

Proposition 3.1. *If there exists a feasible solution for the train scheduling problem, the following conditions have to be satisfied:*

$$\begin{aligned} H_{min} &\le \Pi\, h_{max} \\ H_{max} &\ge \Pi\, h_{min} \end{aligned} \tag{28}$$

Proof. We consider the contrapositive of the proposition: If $H_{min} > \Pi\, h_{max}$ or $H_{max} < \Pi\, h_{min}$, then there is no feasible solution for the train scheduling problem.

Following constraint (27), the first condition can be expanded as:

$$\sigma_{n,0_\pi} - \sigma_{n-k,0_\pi} \ge H_{min} > \Pi\, h_{max}$$

If this condition is holds for $k \le \Pi$, we should have

$$max\{\sigma_{n,0_\pi} - \sigma_{n-k,0_\pi}\} = k\, h_{max} > \Pi\, h_{max}$$

This contradiction arises because $k \le \Pi$ cannot be satisfied. Therefore, there is no feasible solution for $k \le \Pi$.

If the first condition is holds for $k > \Pi$, then there must exist a route π' that satisfies the following condition:

$$T^{\pi'}_{[\sigma_{n-k,0_\pi}\ ,\ \sigma_{n,0_\pi}]} \geq \frac{k-1}{\Pi - 1}$$

during the time interval $[\sigma_{n-k,0_\pi}\ ,\ \sigma_{n,0_\pi}]$. The notation $T^{\pi'}_{[\sigma_{n-k,0_\pi}\ ,\ \sigma_{n,0_\pi}]}$ represents the number of dispatching times on route π' during that time interval.

Furthermore, with the assumption that the time interval $[\sigma_{m-m',0_{\pi'}}\ ,\ \sigma_{m,0_{\pi'}}]$ is the minimum dispatching headway of π' during the time interval $[\sigma_{n-k,0_\pi}\ ,\ \sigma_{n,0_\pi}]$, we can derive the minimum dispatching headway of π' has to satisfy:

$$\begin{aligned}\sigma_{m,0_{\pi'}} - \sigma_{m-m',0_{\pi'}} &\leq m'\, h_{max} \\ &\leq \frac{k-1}{T^{\pi'}_{[\sigma_{n-k,0_\pi}\ ,\ \sigma_{n,0_\pi}]}}\, h_{max} \\ &< \Pi\, h_{max}\end{aligned}$$

This contradicts with $H_{min} > \Pi\, h_{max}$. Therefore, if $H_{min} > \Pi\, h_{max}$, there is no feasible solution for the train scheduling problem.

For the second condition, referring to constraint (27), it can be expanded as follows:

$$\sigma_{n,0_\pi} - \sigma_{n-k,0_\pi} \leq H_{max} < \Pi\, h_{min}$$

Similarly to the proof of the first condition, we can also establish that there is no feasible solution for the train scheduling problem if $H_{max} < \Pi\, h_{min}$.

Therefore, the original proposition has been proven through a contrapositive proof. □

Proposition 3.2. *The sufficient condition for the existence of a feasible solution for the train scheduling problem with respect to constraints (25)–(28) is:*

$$\sigma_{n,0_\pi} \leq H_{max} + \sigma_{n-m,0_{\pi'}} - h_{min} \tag{29}$$

for any $\pi' \neq \pi$. The index $n-m$ represents the last service run that departs on route π' among the past dispatches.

Proof. Let us start with considering the case where $n-m > n-k$ (i.e. $\sigma_{n-m,0_{\pi'}} > \sigma_{n-k,0_\pi}$). Following constraint (27), the dispatching time $\sigma_{n,0_\pi}$ of service run n can be expressed as:

$$\begin{aligned}\sigma_{n,0_\pi} &\leq H_{max} + \sigma_{n-k,0_\pi} \\ &= H_{max} + \sigma_{n-m,0_{\pi'}} - (\sigma_{n-m,0_{\pi'}} - \sigma_{n-k,0_\pi}) \\ &\leq H_{max} + \sigma_{n-m,0_{\pi'}} - h_{min}\end{aligned}$$

Therefore, the constraints will be satisfied if $n-m > n-k$.

For the case $n-m < n-k$, the dispatching time $\sigma_{n+m',0_{\pi'}}$ of the first service run that departs on route π' after service run n should satisfy the following conditions:

$$\begin{aligned}\sigma_{n+m',0_{\pi'}} &\leq H_{max} + \sigma_{n-m,0_{\pi'}} \\ &= H_{max} + \sigma_{n,0_\pi} - \sigma_{n,0_\pi} + \sigma_{n-m,0_{\pi'}}\end{aligned}$$

which can be rearranged as:

$$\begin{aligned}\sigma_{n,0_\pi} &\leq H_{max} + \sigma_{n-m,0_{\pi'}} - (\sigma_{n+m',0_{\pi'}} - \sigma_{n,0_\pi}) \\ &\leq \max\{H_{max} + \sigma_{n-m,0_{\pi'}} - (\sigma_{n+m',0_{\pi'}} - \sigma_{n,0_\pi})\} \\ &= H_{max} + \sigma_{n-m,0_{\pi'}} - h_{min}\end{aligned}$$

Therefore, there exists at least a feasible $\sigma_{n+m',0_{\pi'}}$ satisfy the constraints (25)–(28) as long as $\sigma_{n,0_\pi} \leq H_{max} + \sigma_{n-m,0_{\pi'}} - h_{min}$. □

In addition to the constraints on dispatching headways, there are constraints related to the rolling stock circulation due to the limited availability of driving modules $\hat{G}$ and trailers $\hat{U}$. Let us consider the route π^* with the longest nominal running time. Assuming that service run k is the Kth train service to arrive at the shared terminal station I and is also the first train to reach the terminal station along route π^*. For service run n, where $n \geq \hat{G} + K$, it is essential to ensure that it departs after the $M = (n - \hat{G})$th return service. These conditions can be expressed as Ying et al. (2020) and Ying et al. (2022):

$$\sigma_{n-1,0} + h_n \geq \hat{\sigma}^{n-1}_{M,I} + \Delta_{\mathcal{T}}, \tag{30}$$

where $\Delta_{\mathcal{T}}$ is the slack time required for redeploying the returned train.

Moreover, considering the maximum time interval for routes is H_{max}. For any time interval $[\hat{\sigma}^{n-1}_{M,I}, \hat{\sigma}^{n-1}_{M,I} + H_{max})$, where $n \geq \hat{G} + K$, there will be a minimum of Π train services returning to the terminal station. This implies that by ensuring the availability of at least Π driving modules for the departure of service run $n = \hat{G} + K$, a feasible solution can be achieved that meets the requirements for rolling stock circulation in the transit network. These conditions can be formulated as:

$$\sigma_{n-1,0} + h_n \geq \hat{\sigma}^{n-1}_{K,I} + \Delta_{\mathcal{T}} + H_{max}. \tag{31}$$

Constraint (31) may potentially conflict with the upper bound of constraint (25) if $\sigma_{n-1,0}$ is smaller than $\hat{\sigma}^{n-1}_{K,I} + \Delta_{\mathcal{T}} + H_{max} - h_{max}$. To ensure feasibility when setting h_n, a preventive lower bound constraint on h_n is introduced for $1 \le n \le \hat{G} + K - 1$ as:

$$\sigma_{n-1,0} + h_n \ge \begin{cases} \sigma_{n-1,0} + T_{\pi^*} + \Delta_{\mathcal{T}} - (\hat{G} + K - n)h_{max} + H_{max}, & \text{if } 1 \le n \le k, \\ \sigma_{k,0} + T_{\pi^*} + \Delta_{\mathcal{T}} - (\hat{G} + K - n)h_{max} + H_{max}, & \text{otherwise,} \end{cases} \tag{32}$$

where service run k is the first train to reach the terminal station along route π^*, and parameter T_{π^*} is the total journey time of route π^*. On this basis, the existence of feasible h_n for all $n \ge \hat{G} + K$ is guaranteed.

To investigate the feasibility associated with trailer availability, we denote $\hat{U}^n_t$ as the number of available trailers at time t associated with service run n. This variable $\hat{U}^n_t$ is initialized as $\hat{U}^1_t = \hat{U}$, which represents the initial number of available trailers before any deployment. Given that fleet sizes u_{n-1} are deployed for service run $n-1$ on route π during the time interval $[\sigma_{n-1,0_\pi}, \sigma_{n-1,I\pi} + \Delta_{\mathcal{T}}]$, the number of available trailers $\hat{U}^n_t$ can be updated as Nguyen and Chow (2023):

$$\hat{U}^n_t = \begin{cases} \hat{U}^{n-1}_t - (u_{n-1} - 1), & \text{if } t \in [\sigma_{n-1,0_\pi}, \sigma_{n-1,I_\pi} + \Delta_{\mathcal{T}}], \\ \hat{U}^{n-1}_t, & \text{otherwise.} \end{cases} \tag{33}$$

Therefore, the constraints related to fleet sizes (including driving modules and trailers) for all service runs can be expressed as follows:

$$\begin{aligned} &1 \le u_n \le min\{u_{max}, 1 + \hat{U}^n_{(\sigma_{n-1,0_\pi} + h_n)}\}, \\ &u_n \in \mathbb{Z}^+. \end{aligned} \tag{34}$$

The right-hand side of u_n ensures that the fleet sizes assigned to service run n do not exceed the number of available fleets at the time of dispatch.

Finally, we impose a constraint to address potential track usage conflicts and ensure that a train service cannot enter a station before its predecessor has departed (see Assumption 1). This safety constraint can be expressed as:

$$\hat{\sigma}^n_{M,i_\pi} - w_{i_\pi} \ge \hat{\sigma}^n_{M-1,i_{\pi'}} + \Delta_S, \tag{35}$$

where Δ_S is the nominal safety margin between any service run pair in the transit network.

The formulas (24)–(35) specify the operational constraints considered in this study. All constraints are related to the dispatching times of previous trains, which can be obtained from $\hat{\boldsymbol{\sigma}}_{n-1,i}$ in the current state. Furthermore, the rolling stock circulation-related constraints, represented by formulas (30)–(34), depend on the assigned fleet sizes of previous trains, which can be derived from $(u_0, \dots, u_{n-1})$ in the current state. Therefore, the Markov property is satisfied when these constraints are taken into account.

4. Reinforcement learning solution approach

This section presents a solution approach based on deep reinforcement learning techniques. The notations used in this section are listed in Table 2.

We first derive the solution $\boldsymbol{x}^*_n$ of (23) from the following optimality condition:

$$\boldsymbol{x}^*_n = \arg\max_{\boldsymbol{x}_n} \mathbb{E}_{\boldsymbol{s}_{n+1} \sim \boldsymbol{P}(\boldsymbol{s}_n, \boldsymbol{x}_n)}[-r_n + \gamma V(\boldsymbol{s}_{n+1})], \tag{36}$$

where decision variable $\boldsymbol{x}_n$ is an element of the decision set $\boldsymbol{\Omega}_n = \{\boldsymbol{x}^1_n, \boldsymbol{x}^2_n, \dots, \boldsymbol{x}^D_n\}$, which represents the set of admissible decisions satisfying constraints (24)–(26). The state-value function $V(\boldsymbol{s}_{n+1})$ denotes the optimal cost-to-go from stage $n+1$ to the final stage N, given that the state at stage $n+1$ is $\boldsymbol{s}_{n+1}$. The state-value function evolves according to the established Bellman optimality equation over stages as follows: (Powell, 2011; Bertsekas, 2019):

$$V(\boldsymbol{s}_n) = \max_{\boldsymbol{x}_n} \mathbb{E}_{\boldsymbol{s}_{n+1} \sim \boldsymbol{P}(\boldsymbol{s}_n, \boldsymbol{x}_n)}[-r_n + \gamma V(\boldsymbol{s}_{n+1})]. \tag{37}$$

The Bellman optimality equation can also be expressed in terms of the action-value function as follows (Sutton and Barto, 2018):

$$Q(\boldsymbol{s}_n, \boldsymbol{x}_n) = \mathbb{E}_{\boldsymbol{s}_{n+1} \sim \boldsymbol{P}(\boldsymbol{s}_n, \boldsymbol{x}_n)}[-r_n + \gamma \max_{\boldsymbol{x}_{n+1}} Q(\boldsymbol{s}_{n+1}, \boldsymbol{x}_{n+1})], \tag{38}$$

where $Q(\boldsymbol{s}_n, \boldsymbol{x}_n)$ represents the cumulative expected cost from stage n to the final stage N, given the prevailing states $\boldsymbol{s}_n$ and decisions $\boldsymbol{x}_n$, respectively.

Furthermore, the action-value function can be iteratively learned using the Q-learning (Powell, 2011; Yin et al., 2016):

$$Q(\boldsymbol{s}_n, \boldsymbol{x}_n) \leftarrow Q(\boldsymbol{s}_n, \boldsymbol{x}_n) + \omega[-r_n + \gamma \max_{\boldsymbol{x}_{n+1}} Q(\boldsymbol{s}_{n+1}, \boldsymbol{x}_{n+1}) - Q(\boldsymbol{s}_n, \boldsymbol{x}_n)]. \tag{39}$$

Computing the exact value of $V(\boldsymbol{s}_n)$ or $Q(\boldsymbol{s}_n, \boldsymbol{x}_n)$ in large state or action spaces is an extremely computationally demanding task. This challenge is commonly referred to as the curse of dimensionality (Powell, 2011; Sutton and Barto, 2018; Bertsekas, 2019).

Moreover, in a partially observable environment, direct access to the state $\boldsymbol{s}_n$ is not available. Instead, the controller can only access the observation $\boldsymbol{o}_n$, which is generated from the underlying state $\boldsymbol{s}_n$. As a straightforward extension of Q-learning in partially observable environment, the iterative formula in (39) can be expressed as:

$$Q(\boldsymbol{o}_n, \boldsymbol{x}_n) \leftarrow Q(\boldsymbol{o}_n, \boldsymbol{x}_n) + \omega[-r_n + \gamma \max_{\boldsymbol{x}_{n+1}} Q(\boldsymbol{o}_{n+1}, \boldsymbol{x}_{n+1}) - Q(\boldsymbol{o}_n, \boldsymbol{x}_n)]. \tag{40}$$

Table 2
Notations used in the reinforcement learning solution approach.

Notation	Definition
Indices	
e	Index of training epoch.
b	Index of transition sequence.
Sets	
$\mathcal{Z}$	Set of replay buffer.
$\boldsymbol{\Omega}_n$	Set of admissible decisions.
Vectors	
$\boldsymbol{\theta}, \boldsymbol{\theta}'$	Parameter vectors of ANN surrogate.
$\hbar_n$	Hidden state of the LSTM layer.
$\tilde{\boldsymbol{h}}_n$	Output vector of the LSTM layer.
$\tilde{\boldsymbol{c}}_n, \boldsymbol{c}_n$	Memory vectors of the LSTM layer.
$\tilde{\boldsymbol{G}}_n^f, \tilde{\boldsymbol{G}}_n^i, \tilde{\boldsymbol{G}}_n^o$	Signal vectors of the LSTM layer.
$\boldsymbol{U}^f, \boldsymbol{U}^i, \boldsymbol{U}^o, \boldsymbol{U}^c$	Underlying parameter vectors of the LSTM layer.
$\boldsymbol{W}^f, \boldsymbol{W}^i, \boldsymbol{W}^o, \boldsymbol{W}^c$	Underlying parameter vectors of the LSTM layer.
Parameters	
Z	Size of replay buffer.
E	Maximum length of training epoch.
L	Length of each transition sequence.
B	Batch size of transition sequences drawn from the replay buffer.
ξ	Smoothing factor for updating parameters of ANN surrogate.
ϵ_{decay}	Parameters to determine the exploration rate.
$\epsilon_{min}, \epsilon_{max}$	Minimum and maximum value of exploration rate.
Environment variables	
Y_b	Target value for updating parameters of ANN surrogate.
$\boldsymbol{x}_n^d$	Elements realized from $\boldsymbol{\Omega}_n$.
$\bar{\boldsymbol{x}}_n$	Selected decision at stage n.
ϵ_e	Exploration rate.
Functions	
$V(\boldsymbol{s}_n)$	State-value function associated with $\boldsymbol{s}_n$.
$Q(\boldsymbol{s}_n, \boldsymbol{x}_n)$	Action-value function associated with $\boldsymbol{s}_n$ and $\boldsymbol{x}_n$.
$Q(\boldsymbol{o}_n, \hbar_n, \boldsymbol{x}_n)$	Action-value function associated with $\boldsymbol{o}_n$, $\hbar_n$ and $\boldsymbol{x}_n$.
$\bar{Q}_{\boldsymbol{\theta}}(\boldsymbol{o}_n, \hbar_n, \boldsymbol{x}_n)$	Masked action-value function associated with $\boldsymbol{o}_n$, $\hbar_n$ and $\boldsymbol{x}_n$.
$\mathcal{L}_{\bar{Q}}(\boldsymbol{\theta})$	Loss function for determining parameters of ANN surrogate.

This can lead to bad estimation of the action-value function, since $Q(\boldsymbol{s}_n, \boldsymbol{x}_n) \neq Q(\boldsymbol{o}_n, \boldsymbol{x}_n)$ (Hausknecht and Stone, 2015). The main challenge in estimating the action-value function in a partially observable environment is the discrepancy between $Q(\boldsymbol{s}_n, \boldsymbol{x}_n)$ and $Q(\boldsymbol{o}_n, \boldsymbol{x}_n)$ (Hausknecht and Stone, 2015; Zhu et al., 2017).

To address the computational challenge and reduce the discrepancy between $Q(\boldsymbol{s}_n, \boldsymbol{x}_n)$ and $Q(\boldsymbol{o}_n, \boldsymbol{x}_n)$, this study proposes an artificial neural network (ANN) surrogate to approximate the action-value (also known as the Q-value). In the following subsections, we first introduce the structure of the ANN surrogate, incorporating operational constraints. We then present the deep recurrent Q network (DRQN) algorithm as the training method for this surrogate.

4.1. Action-value function approximation

Fig. 4 depicts the structure of the ANN surrogate, denoted as $Q_{\boldsymbol{\theta}}$, which is defined by a parameter set $\boldsymbol{\theta}$. The surrogate consists of an LSTM (long short-term memory) layer (Hochreiter and Schmidhuber, 1997; Zhuang et al., 2025; Wu et al., 2025b,a), and two fully connected layers with element-wise non-linear activation function (e.g., hyperbolic tangent (Tanh) and rectified linear unit (ReLU) function Goodfellow et al., 2016; Murphy, 2022). The inclusion of an LSTM layer in the framework allows for the integration of a long history of observations, utilizing $Q(\boldsymbol{o}_n, \hbar_n, \boldsymbol{x}_n)$ instead of $Q(\boldsymbol{o}_n, \boldsymbol{x}_n)$, thus enabling more accurate estimation of the Q-value (Hausknecht and Stone, 2015). At each stage n, given the observation $\boldsymbol{o}_n$ and the hidden state $\hbar_n = (\tilde{\boldsymbol{h}}_n, \tilde{\boldsymbol{c}}_n)$, the LSTM layer generates a new hidden state $\hbar_{n+1} = (\tilde{\boldsymbol{h}}_{n+1}, \tilde{\boldsymbol{c}}_{n+1})$, comprising the output vector $\tilde{\boldsymbol{h}}_{n+1}$ and the memory vector $\tilde{\boldsymbol{c}}_{n+1}$. The output vector $\tilde{\boldsymbol{h}}_{n+1}$ is then utilized as input for the fully connected layers to estimate the Q-value for each decision $\boldsymbol{x}_n$.

Fig. 5 further introduces the detailed structure of the LSTM layer. Given the observation $\boldsymbol{o}_n$ and the output vector $\tilde{\boldsymbol{h}}_n$, the variables in the LSTM layer evolve according to the following equations (Goodfellow et al., 2016; Zhuang et al., 2025):

$$\begin{aligned}
&\text{(forget gate)} && \tilde{\boldsymbol{G}}_n^f = \mathrm{Sigm}(\boldsymbol{W}^f \boldsymbol{o}_n + \boldsymbol{U}^f \tilde{\boldsymbol{h}}_n),\\
&\text{(input gate)} && \tilde{\boldsymbol{G}}_n^i = \mathrm{Sigm}(\boldsymbol{W}^i \boldsymbol{o}_n + \boldsymbol{U}^i \tilde{\boldsymbol{h}}_n),\\
&\text{(output gate)} && \tilde{\boldsymbol{G}}_n^o = \mathrm{Sigm}(\boldsymbol{W}^o \boldsymbol{o}_n + \boldsymbol{U}^o \tilde{\boldsymbol{h}}_n),\\
&\text{(memory vector)} && \boldsymbol{c}_{n+1} = \mathrm{Tanh}(\boldsymbol{W}^c \boldsymbol{o}_n + \boldsymbol{U}^c \tilde{\boldsymbol{h}}_n),
\end{aligned} \tag{41}$$

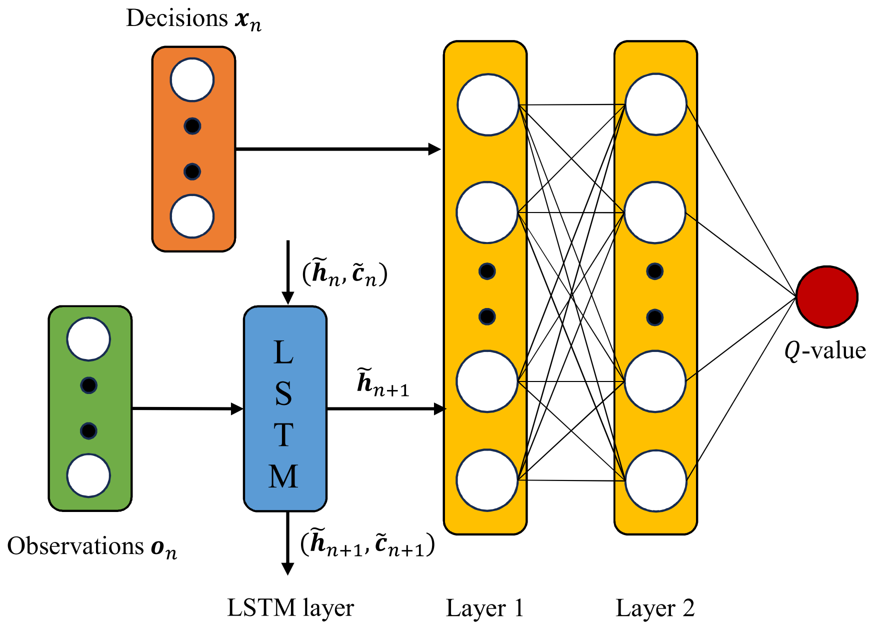


Fig. 4. Structure of the ANN surrogate for action-value function approximation.

where Sigm and Tanh represent the sigmoid and hyperbolic tangent function, respectively. The variables $\boldsymbol{W}^f, \boldsymbol{W}^i, \boldsymbol{W}^o$, and $\boldsymbol{W}^c$, as well as $\boldsymbol{U}^f, \boldsymbol{U}^i, \boldsymbol{U}^o$, and $\boldsymbol{U}^c$, are the underlying parameters that determine the LSTM layer signals $\tilde{\boldsymbol{G}}_n^f, \tilde{\boldsymbol{G}}_n^i, \tilde{\boldsymbol{G}}_n^o$, and the memory vector $\boldsymbol{c}_{n+1}$.

Given the memory vector $\tilde{\boldsymbol{c}}_n$ and $\boldsymbol{c}_{n+1}$, the new memory vector $\tilde{\boldsymbol{c}}_{n+1}$ of the LSTM layer can be calculated as:

$$\tilde{\boldsymbol{c}}_{n+1} = \tilde{\boldsymbol{G}}_n^f \odot \tilde{\boldsymbol{c}}_n + \tilde{\boldsymbol{G}}_n^i \odot \boldsymbol{c}_{n+1}, \tag{42}$$

where the notation $\odot$ means the element-wise product of the two matrices. Furthermore, the new output vector $\tilde{\boldsymbol{h}}_{n+1}$ of the LSTM layer can be updated as:

$$\tilde{\boldsymbol{h}}_{n+1} = \tilde{\boldsymbol{G}}_n^o \odot \mathrm{Tanh}(\tilde{\boldsymbol{c}}_{n+1}), \tag{43}$$

The estimated Q-values for the decision set $\{\boldsymbol{x}_n^1, \boldsymbol{x}_n^2, \dots, \boldsymbol{x}_n^D\}$ are then generate based on the output vector $\tilde{\boldsymbol{h}}_{n+1}$ and the fully connected layers. Nevertheless, the running process does not consider explicitly the stage-dependent constraints (24)–(35). To incorporate the stage-dependent constraints imposed on the decision set, we also employ invalid action masking (IAM) as illustrated below:

$$\bar{Q}_\theta(\boldsymbol{o}_n, \hbar_n, \boldsymbol{x}_n^d) = \begin{cases} Q_\theta(\boldsymbol{o}_n, \hbar_n, \boldsymbol{x}_n^d), & \text{if } \boldsymbol{x}_n^d \text{ meets constraints (24)–(35)}, \\ -\infty, & \text{otherwise}, \end{cases} \tag{44}$$

for $d = 1, 2, \dots, D$. In cases where decisions do not adhere to constraints (24)–(35), the IAM method can assign a Q-value of $-\infty$ to them. During action selection, only actions with Q-values not equal to $-\infty$ are considered, thus preventing the selection of invalid actions. This masking method has been shown to be effective in value function approximation and policy gradient methods, even in scenarios with large decision spaces (Huang and Ontañón, 2020; Long et al., 2022).

4.2. Training algorithm

Before the proposed train scheduling framework can achieve satisfactory performance, it is necessary to determine the parameters θ of the ANN surrogate. This study uses a deep recurrent Q network (DRQN) training algorithm (Hausknecht and Stone, 2015) due to its computational efficiency in partially observable environments. The procedure of the DRQN is summarized in algorithm 1, with further details provided below.

The training process starts with creating a replay buffer $\mathcal{Z}$, which is designed to store a queue of size Z for storing observed transition samples $(\boldsymbol{o}_n, \hbar_n, \bar{\boldsymbol{x}}_n, r_n, \boldsymbol{o}_{n+1}, \eta_n)$ between successive stage pairs across simulation runs. The replay buffer is a dynamic library of ground truths for training the ANN surrogate throughout the training epochs, denoted as $e = 1, 2, \dots, E$. To enhance stability and promote convergence in the training process, we introduce a target ANN surrogate with parameters θ' (Lillicrap et al., 2015; Mnih et al., 2015), which are initialized as: $\theta' \leftarrow \theta$.

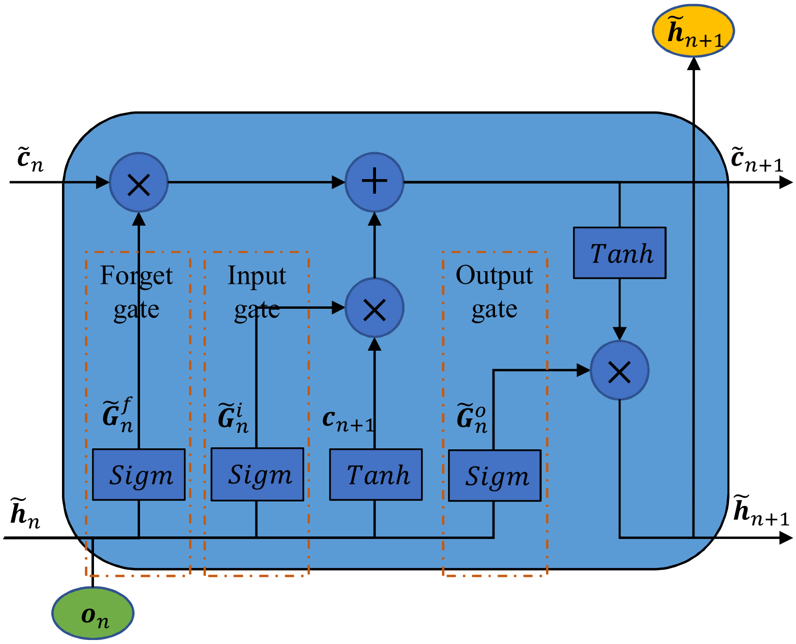


Fig. 5. Structure of the LSTM layer.

Algorithm 1 Deep recurrent Q network (DRQN) training algorithm

Input: Initial ANN surrogate parameters $\boldsymbol{\theta}$
Output: Trained ANN surrogate parameters for generating feasible decisions that maximize (23)
Initialize the target ANN surrogate parameters: $\boldsymbol{\theta}' \leftarrow \boldsymbol{\theta}$
Initialize the replay buffer with Z transitions randomly chosen from feasible decisions
for $e \leftarrow 1$ to E **do**
    Initialize stage index: $n \leftarrow 0$
    Initialize the MDP with $\boldsymbol{o}_0 \leftarrow \mathbf{0}$, $\hbar_0 \leftarrow \mathbf{0}$, $\eta_0 \leftarrow 0$
    while $\eta_n = 0$ **do**
        Get the feasible decision $\bar{\boldsymbol{x}}_n$ following the ϵ-greedy strategy
        Execute $\bar{\boldsymbol{x}}_n$ for the corresponding r_n, $\boldsymbol{o}_{n+1}$, and η_n via the transition function $\boldsymbol{P}$
        Store the acquired transition experience $(\boldsymbol{o}_n, \hbar_n, \bar{\boldsymbol{x}}_n, r_n, \boldsymbol{o}_{n+1}, \eta_n)$ into the replay buffer
        Sample a batch of B transition sequences from the replay buffer, where each transition sequence consists of L successive transition experiences
        Calculate the target value for sampled transition sequences $(\boldsymbol{o}_b, \hbar_b, \bar{\boldsymbol{x}}_b, r_b, \boldsymbol{o}_{b+1}, \eta_b)$, $b = 1, 2, \ldots, B$:

$$Y_b = r_b + \gamma(1 - \eta_b) \max \bar{Q}_{\boldsymbol{\theta}'}(\boldsymbol{o}_{b+1}, \hbar_{b+1}, \boldsymbol{x}_{b+1})$$

        Determine new $\boldsymbol{\theta}$ for the ANN surrogate by minimizing the loss:

$$\boldsymbol{\theta} = \arg\min_{\boldsymbol{\theta}} \mathcal{L}_{\bar{Q}}(\boldsymbol{\theta}) = \frac{1}{B} \sum_b (\bar{Q}_{\boldsymbol{\theta}}(\boldsymbol{o}_b, \hbar_b, \boldsymbol{x}_b) - Y_b)^2$$

        Update the target parameters of the ANN surrogate:

$$\boldsymbol{\theta}' \leftarrow \xi\boldsymbol{\theta} + (1 - \xi)\boldsymbol{\theta}'$$

        Update the stage index: $n \leftarrow n + 1$
    end while
    Update the exploration rate: $\epsilon_e = \epsilon_{min} + (\epsilon_{max} - \epsilon_{min}) \cdot (\exp(-e \cdot \epsilon_{decay}))$
end for

Given the prevailing observation o_n and hidden state $\hbar_n$, the corresponding decision $\bar{\boldsymbol{x}}_n$ is derived from its associated masked action-value function $\bar{Q}_{\boldsymbol{\theta}}(\boldsymbol{o}_n, \hbar_n, \boldsymbol{x}_n)$. To prevent the model from getting stuck in local optima, we employ an adaptive ϵ-greedy strategy to determine the decision (Sutton and Barto, 2018; Chow et al., 2021). Specifically, at each decision stage, the controller has a probability of $1-\epsilon$ to select the decision with the highest Q-value as:

$$\bar{\boldsymbol{x}}_n = \arg\max_{\boldsymbol{x}_n} \bar{Q}_\theta(\boldsymbol{o}_n, \hbar_n, \boldsymbol{x}_n). \tag{45}$$

In the remaining cases, a decision is randomly selected from those with Q-values not equal to $-\infty$.

The symbol ϵ in adaptive ϵ-greedy strategy represents the exploration rate, determining the balance between exploration and exploitation during training. In this study, the exploration rate ϵ_e is set to gradually decrease as the training epoch e progresses to ensure convergence. The update formula for the exploration rate is as follows (Long et al., 2022):

$$\epsilon_e = \epsilon_{min} + (\epsilon_{max} - \epsilon_{min}) \cdot (\exp(-e \cdot \epsilon_{decay})), \tag{46}$$

where ϵ_{min} and ϵ_{max} represent the lower and upper bounds of the exploration rate, respectively. The decay factor ϵ_{decay} controls the rate at which the exploration rate decreases as the training epoch e progresses.

The selected decision is executed for the corresponding reward r_n, observation $\boldsymbol{o}_{n+1}$, and terminal indicator η_n. The transition experience $(\boldsymbol{o}_n, \hbar_n, \bar{\boldsymbol{x}}_n, r_n, \boldsymbol{o}_{n+1}, \eta_n)$ is then stored into and update the replay buffer queue.

The bootstrapped random updates strategy (Hausknecht and Stone, 2015) is employed in this study to sample transition sequences from the replay buffer. This strategy is chosen because it effectively utilizes the hidden state of the LSTM layer (Hausknecht and Stone, 2015; Kapturowski et al., 2018). Specifically, a batch of B transition sequences is sampled from the updated replay buffer, where each sequence consists of L successive transition experiences. The target value Y_b associated with each batch of transition sequences $(\boldsymbol{o}_b, \hbar_b, \bar{\boldsymbol{x}}_b, r_b, \boldsymbol{o}_{b+1}, \eta_b)$ can then be calculated by one-step temporal difference (TD) learning principle (Sutton and Barto, 2018):

$$Y_b = r_b + \gamma(1-\eta_b) \max \bar{Q}_{\theta'}(\boldsymbol{o}_{b+1}, \hbar_{b+1}, \boldsymbol{x}_{b+1}) \tag{47}$$

for all $b = 1, 2, \ldots, B$.

The ANN surrogate parameters are then updated via the following stochastic batch optimization method (Goodfellow et al., 2016), which aims to find new θ that minimize the loss function $\mathcal{L}_{\bar{Q}}$ between each target value Y_b and the corresponding Q-values in the batch:

$$\theta = \arg\min_\theta \mathcal{L}_{\bar{Q}}(\theta) = \frac{1}{B}\sum_b (\bar{Q}_\theta(\boldsymbol{o}_b, \hbar_b, \boldsymbol{x}_b) - Y_b)^2. \tag{48}$$

The optimization problems described above can be effectively solved by many algorithms, such as the well-established Adam algorithm (Kingma and Ba, 2014), which will be used in this study.

For convergence and stability of the training process (Mnih et al., 2015), the target parameters in the ANN surrogate will be updated using a moving averaging technique:

$$\theta' \leftarrow \xi\theta + (1-\xi)\theta', \tag{49}$$

where ξ is a sufficiently small smoothing factor.

The training process will be repeated until a predefined number of iterations E has been reached. When the training of the ANN surrogate is complete, the trained controller or surrogate can generate feasible decisions with satisfactory performance in real-time, regardless of the scale of the problem. Specifically, the train scheduling system is first characterized by the state $\boldsymbol{s_n}$. Then, the controller receives an observation $\boldsymbol{o_n}$ based on the state $\boldsymbol{s_n}$ and generates a feasible decision $\boldsymbol{x_n}$ in real-time, including dispatching headway, dispatching route, and assigned fleet sizes, based on the current observation and related constraints. Subsequently, the system transitions from $\boldsymbol{s_n}$ to $\boldsymbol{s_{n+1}}$ with the decision $\boldsymbol{x_n}$ through the transition function $\boldsymbol{P}$. This process repeats in a loop until all passengers have been served.

It should be noted that the proposed framework in this study is highly scalable and can be easily extended to accommodate more complex transit network settings. Specifically, when considering more complex transit networks (such as additional stations, more dispatching routes, and larger assigned fleet sizes for service runs), the operational constraints imposed on the simulation system remain valid regardless of the complexity of the network or the number of trains. The necessary adjustments involve reconstructing the simulation system and modifying the state space, observation space, action space, and training hyperparameters accordingly. While these changes may increase the offline training time of the surrogate, they do not affect its convergence or online implementation.

5. Case study

5.1. Experiment settings

The proposed optimization framework is tested on three routes of the Hong Kong Light Rail Transit (LRT) network, which consists of 42 stations in both outbound and inbound directions. Service runs in the transit network will depart from station 001, follow the predefined route towards station 600, and then return to station 001. The total number of driving modules $\hat{G}$ and trailers $\hat{U}$ is 42 and 16, respectively. Additionally, each driving module or trailer is capable of carrying up to $\hat{C} = 200$ passengers. The test covers five hours from 5:00 to 10:00 on a typical weekday.

Fig. 7(a) shows the spatial distribution of passenger demand over the network, while Fig. 7(b) shows the temporal variations of total passenger demand. These data are derived from anonymous automatic fare (Octopus) transaction records. As mentioned in

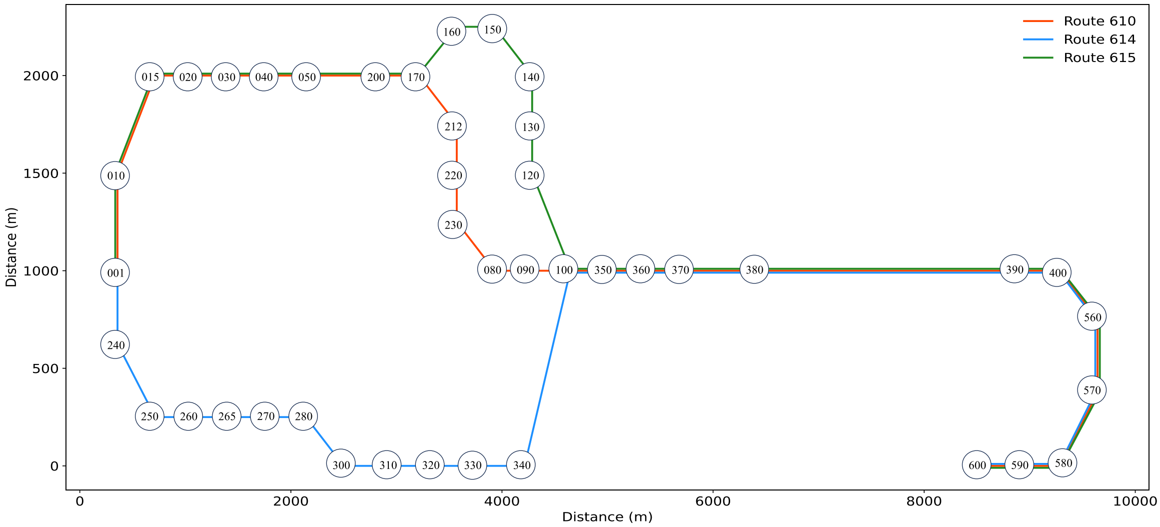


Fig. 6. Configuration of Hong Kong Light Rail network.

our previous studies (Nguyen et al., 2021; Nguyen and Chow, 2023), we are obliged to adhere to the request of the Mass Transit Railway Corporation (MTR), and we can only display the relative demand level rather than the actual demand values.

According to the MTR operational guidelines, the safety separation between successive train services Δ_{S} and the slack time $\Delta_{\mathcal{T}}$ are both set to 60 s, and the nominal dwell time w_i at each station is set to 30 s. Additionally, the nominal journey times for routes 610, 614, and 615 are specified as 96 min, 93 min, and 100 min, respectively. There are five admissible levels of dispatching headways, which are 2, 4, 6, 8, and 10 min, with corresponding $h_{min} = 2$ and $h_{max} = 10$. Furthermore, the minimum headway H_{min} and maximum headway H_{max} for each route are set to 2 and 30 min, respectively. Considering track characteristics, the maximum fleet sizes for each service run are restricted to 2 (i.e., $u_{max} = 2$). Thus, the observation space in this problem is represented by a feature vector of size 168 (= 84 (the remaining passengers at each station) + 84 (the last departure times at each station). The action space is characterized by a feature vector of size 30 (=5 (dispatching headways) * 2 (fleet sizes) * 3 (number of routes)).

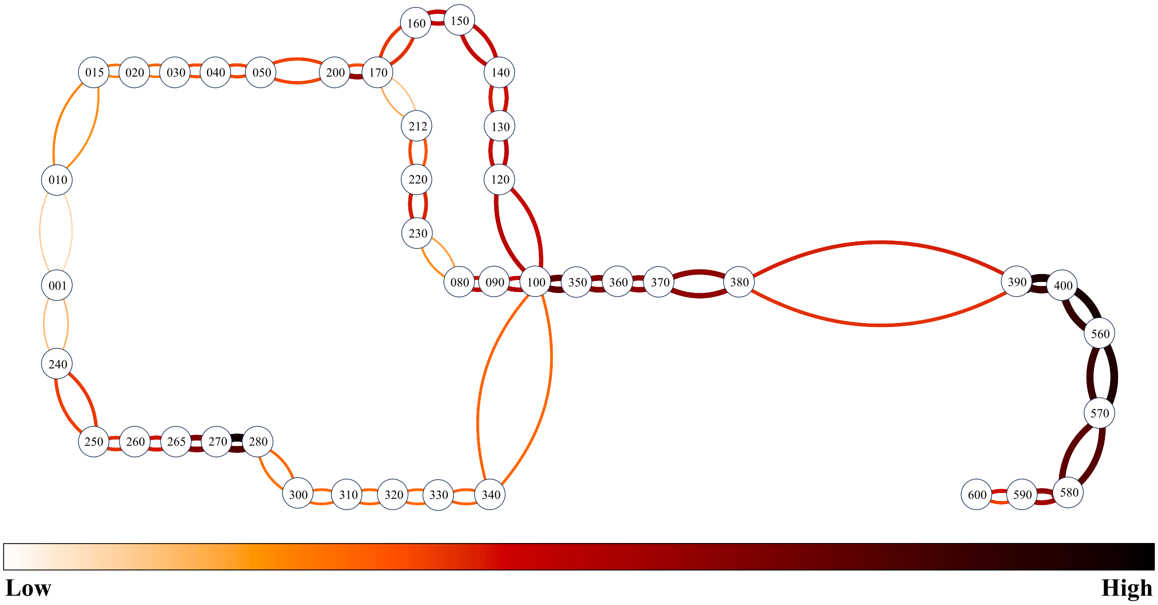
For the coefficients in the reward signals, we first set the trade-off parameter in (22) as $\alpha = 0.5$. We further set the unit monetary cost of passenger waiting and operating as $\zeta^W = 2.61 \times 10^{-2}\$/s$, $\zeta^F = 1.08\$/kWh$, and $\zeta^D = 3.05 \times 10^{-3}\$/m$, respectively. To specify the Davis formula, we set the tare mass of a train unit as $M^{train} = 30.30$ metric tons and the average weight of a passenger with personal belongings as $M^{psg} = 0.08$ metric tons (=80 kg). The coefficients of the Davis function are set as $\beta_1 = 16.600$ kN/ton, $\beta_2 = 0.366$ kN/ton(m/s), and $\beta_3 = 0.026$ kN/ton(m/s)^2.

In the artificial neural network, the hidden state size of the LSTM layer is 32, and the two-layer fully connected layer consists of 512 and 256 neurons, respectively. The ReLU activation function is applied between each fully connected layer to enhance the model's performance. The ANN surrogate is trained using the DRQN algorithm introduced in Section 4.2. The experience replay buffer has a size of $Z = 20000$, and the initial transition experiences are generated using a random policy. During the training process, the batch size for sampling transition sequences from the replay buffer is set to $B = 32$, and the length of successive transition experiences in each sampled transition sequence is set to $L = 10$. The discount factor γ used for calculating the Q-value is set to 0.995. At the end of each training epoch, the ANN surrogate parameters are updated using a smoothing factor of $\xi = 5 \times 10^{-3}$. The total number of training epochs is set to $E = 10000$. Throughout the training process, the exploration rate in the ϵ-greedy strategy gradually decays from the maximum value of $\epsilon_{max} = 1$ to $\epsilon_{min} = 0.1$. The decay rate ϵ_{decay} is set to be 8×10^{-4}. Fig. 8 provides a profile of the exploration rate throughout the training epochs. All experiments are implemented in Python and executed on a desktop computer with an Intel Core i7 3.00 GHz CPU and 16 GB RAM.

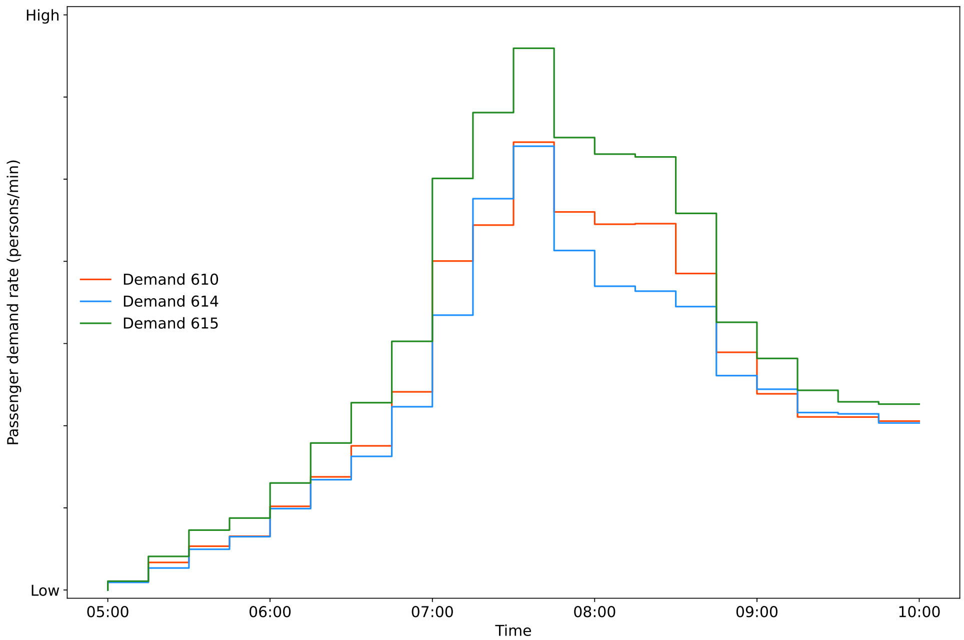
5.2. Training

Fig. 9(a) first shows the training process of the DRQN algorithm. In the figure, the gray line represents the actual performance scores, which are determined by the value of the objective function (i.e., total cost) achieved by the controller in each epoch. The black line represents the moving average performance scores over the last 100 epochs. At the beginning of the training, the controller makes random decisions due to the high exploration rate, resulting in significant fluctuations in performance scores. As the training progresses, around 4000 epochs, the performance score gradually stabilizes at a relatively low level. Fig. 9(b) then shows the best (highest) performance score achieved during the training process. It is observed that the highest performance score of −589528 is attained at approximately epoch 5300.

Fig. 10 further presents the dispatching headways and fleet sizes for each route at the highest achieved performance score. In the figure, the 'point' and 'square' markers represent dispatches of 'single-car' and 'double-car' trains, respectively. The figure



(a) Spatial distribution of passenger demand



(b) Passenger demand profile

Fig. 7. Passenger demand in the test network.

also includes the average passenger demand profile corresponding to each route as a reference. It is noted that in the considered transit network, the total journey time for any given route exceeds 90 min. Therefore, the controller needs to adjust the dispatching headways and fleet sizes in advance to avoid service resource shortages or wastage during service operations.

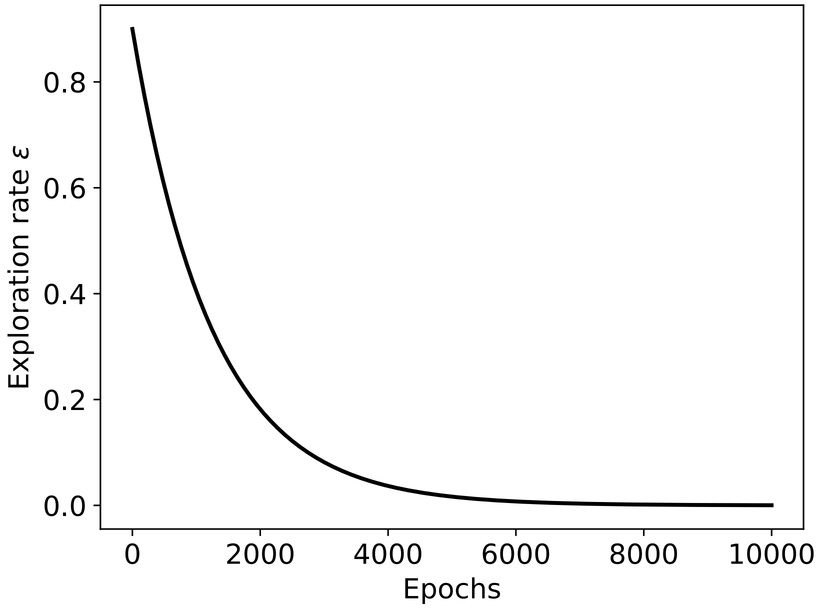


Fig. 8. The exploration rate profile over the training epoch.

Table 3
Performance comparison of different training algorithms.

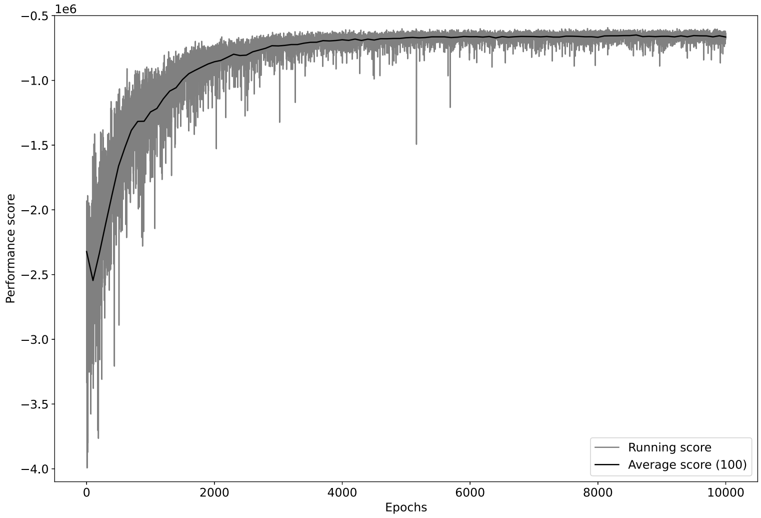
Algorithm	DRQN	DQN	GA	PSO
Performance score	−589528	−621322	−699645	−731467
Differences	–	−5.4%	−18.6%	−24.1%

Specifically, in Fig. 10, it can be observed that the controller has higher average fleet sizes with shorter dispatching headways between 6:00 and 8:00 to cater to the potential peak passenger demand between 7:00 and 9:00. Furthermore, there is a significant difference in the dispatching patterns on route 610 compared to routes 614 and 615 during 6:00 and 8:00. This discrepancy occurs because increasing the frequency on route 615 serves the majority of passenger demand for route 610 due to significant overlapping areas between the two routes (see Fig. 6). Fig. 10 also shows the controller maintains relatively stable dispatching headways during the periods of 8:00 to 9:00 to meet the passenger demand during the off-peak period (i.e. after 9:00). After 9:00, taking advantage of the remaining service resources in the transit network, the controller dispatches service runs with longer headways. This strategy helps minimize operating costs without significantly decreasing passenger satisfaction, showing the adaptiveness of the train scheduling framework to the prevailing passenger demand.

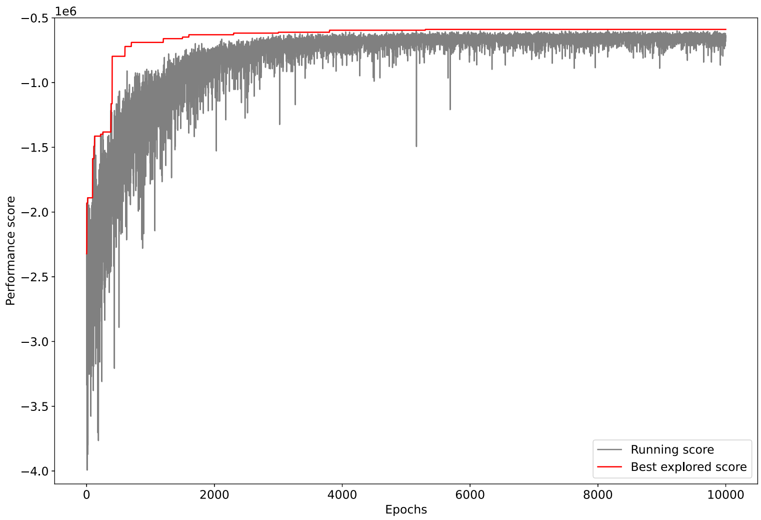
5.3. Benchmarking with other training algorithms

For benchmarking purposes, we compare the proposed DRQN algorithm with the well-established genetic algorithm (GA) and particle swarm optimization (PSO). These benchmarking algorithms are chosen due to their popularity in sophisticated service scheduling problems (Niu and Zhou, 2013; Sun et al., 2014b; Ying et al., 2020; Chow et al., 2024). To investigate the impact of historical observations, we also employ the deep Q network (DQN) algorithm (Mnih et al., 2015) to train a neural network without the LSTM layer. This design represents a situation where the value function evaluation only depends on the current observation without considering historical observations. It is noted that GA and PSO are designed as offline optimization methods, which differ from the intended application of the proposed reinforcement learning framework, an online algorithm. However, in this comparison, the focus is not on performance or applications but on the training process.

Fig. 11 shows the progression of all training algorithms. Furthermore, Table 3 shows the best performance achieved under different training algorithms. It is observed that the reinforcement learning algorithms (DRQN and DQN) outperform GA and PSO in terms of performance score and computation time. That can be attributed to the approximation of the state space based on the ANN surrogate, which is specifically tailored to the original service routing and scheduling problem. As a result, specialized optimization techniques such as the ADAM algorithm can be employed for the training process instead of relying on less efficient meta-heuristic algorithms like GA and PSO. Compared to the DQN algorithm, the DRQN algorithm requires a longer computation time to converge. However, due to its more sophisticated design, it can also deliver better performance scores. The advantage of DRQN over DQN further reveals the effectiveness of the LSTM layer in the partially observable environment. Nevertheless, it is observed that the highest performance score achieved by the DRQN algorithm is not significantly different from that of the simpler



(a) The evolution of average score



(b) The evolution of best score

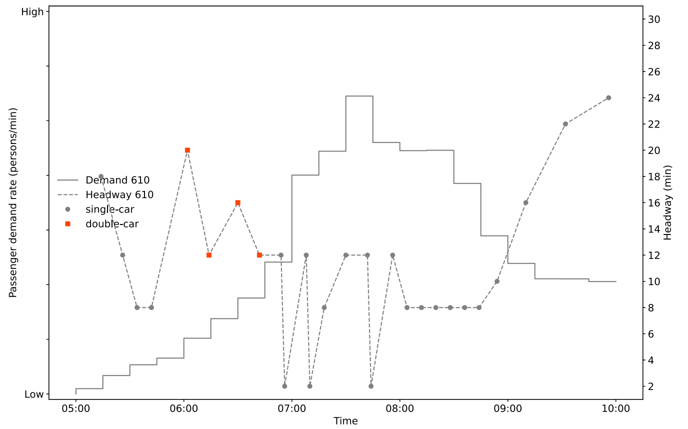
Fig. 9. Training process of the DRQN algorithm.

DQN algorithm. Hence, we may still be able to conclude that both training algorithms are suitable for application in the present case.

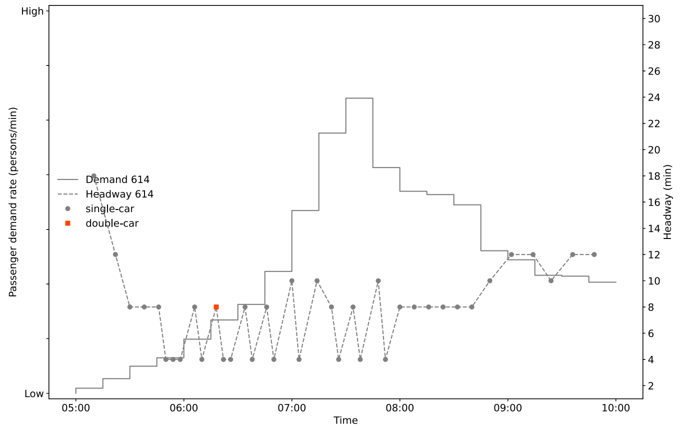
5.4. Comparison with other scheduling strategies

To evaluate the effectiveness of the proposed flexible routing and fleet size strategies, seven scenarios (S1-S7) were constructed and compared (Table 4). S1 incorporates both flexible routing and fleet size strategies, while S2 maintains flexible fleet sizes but follows a fixed dispatching sequence (...-615-614-610-...). S3 and S5 implement flexible routing with double-car and single-car trains, respectively, while S4 and S6 implement fixed routing (as in S2) with double-car and single-car trains, respectively. Additionally, S7 is constructed based on the existing scheduling strategy in practice for comparison.

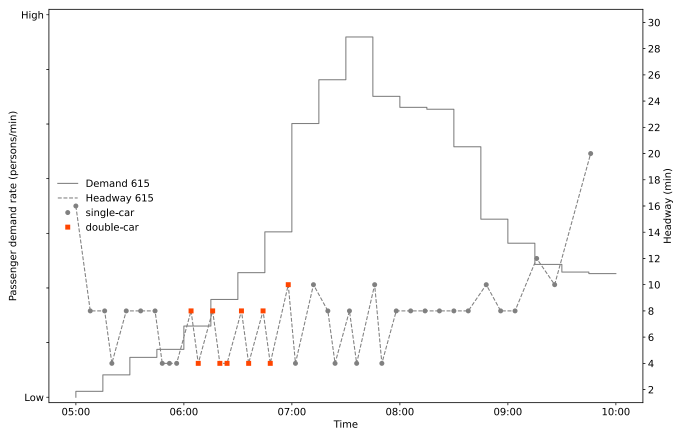
Table 5 summarizes the key performance achieved under different scenarios. The results first demonstrate that the model of S1 delivers the best performance score. Specifically, compared to S2, the model of S1 achieved reductions of 12.1% in passenger cost



(a) Route 610



(b) Route 614



(c) Route 615

Fig. 10. Dispatching headways and fleet sizes with respect to passenger demand profile.

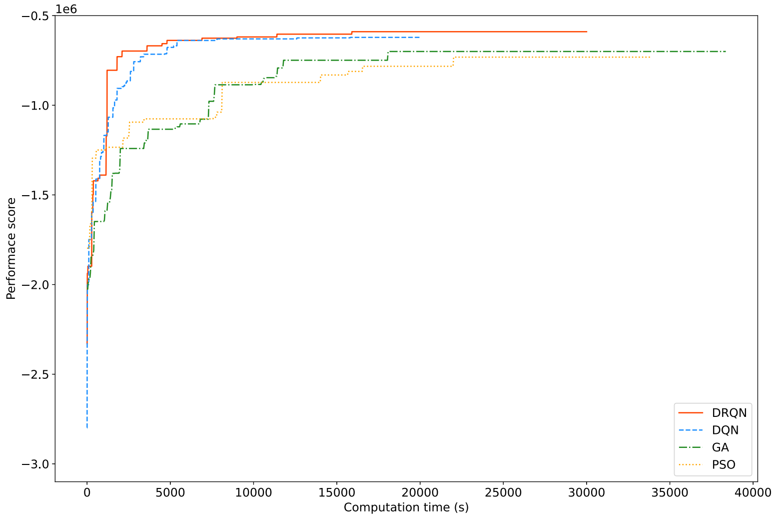


Fig. 11. Progression of training algorithms.

Table 4
Experiment scenarios with different settings.

Experiment	Flexible fleet sizes	Flexible routes
S1	Yes	Yes
S2	Yes	No
S3	Only double-car trains	Yes
S4	Only double-car trains	No
S5	Only single-car trains	Yes
S6	Only single-car trains	No
S7	In practice	In practice

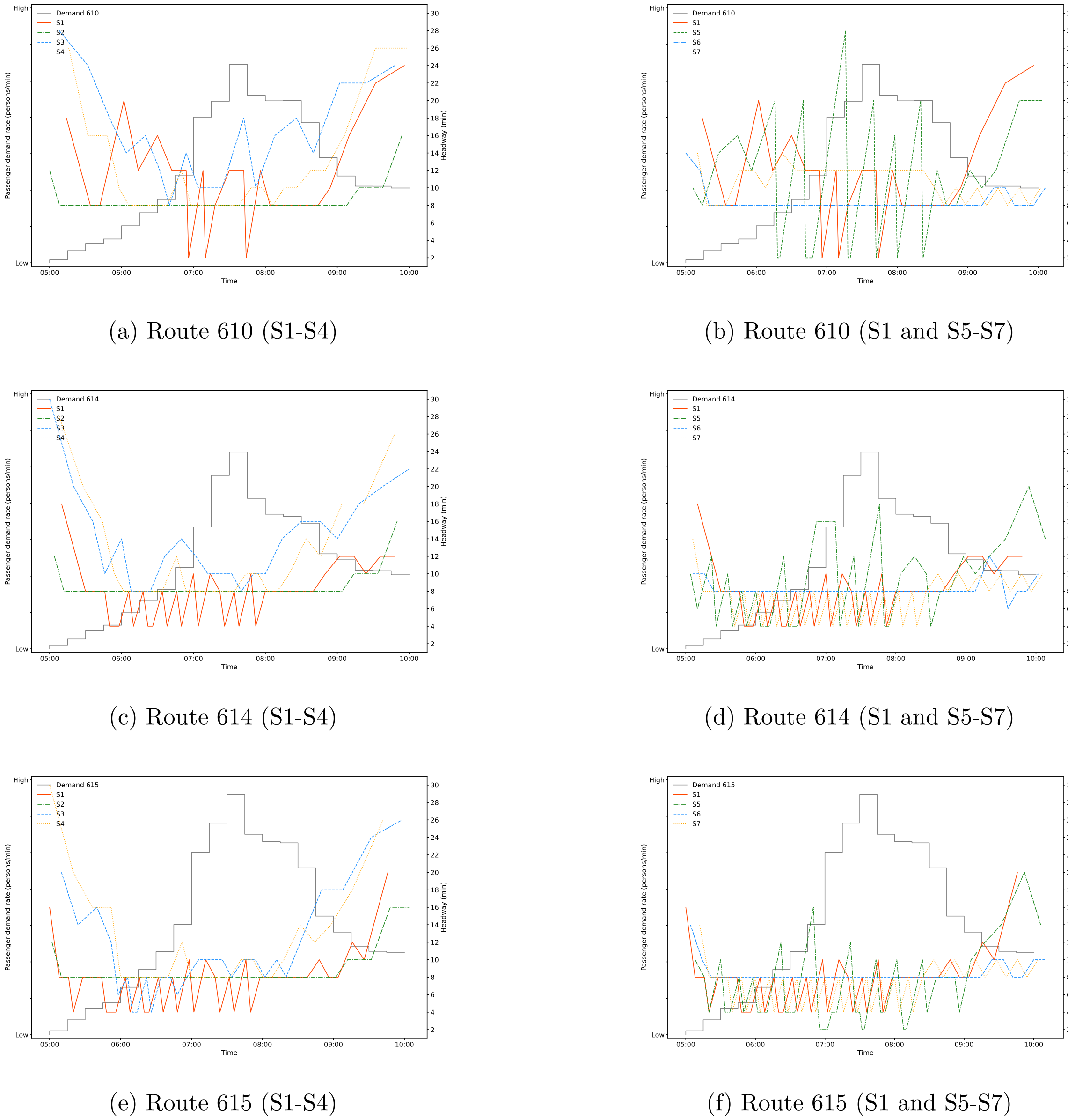
Table 5
Performances of different experiment scenarios.

Experiment	Performance score	Passenger cost ($)	Operating cost ($)	Number of service runs	Average fleet sizes
S1	−589528	401233	777822	105	1.14
S2	−637240	449847	824633	107	1.15
S3	−698349	509834	886864	71	2.00
S4	−735519	502613	968425	80	2.00
S5	−712646	675816	749475	113	1.00
S6	−963759	1157943	769576	114	1.00
S7	−627802	413881	841723	113	1.14

and 6.0% in operating cost. These reductions could be due to the flexibility of scenario S1, which allows increased service runs on highly demanded routes during peak periods and decreased service runs during off-peak periods. This is justified by comparing the dispatching headways depicted in Fig. 12, which shows the dispatching headways under different scenarios. Furthermore, Fig. 13 illustrates the dispatching fleet sizes under different scenarios. It can be observed from Fig. 13 that the models of S1 and S2 dispatch single-car trains during off-peak periods to reduce operating costs and dispatch double-car trains during peak periods to satisfy oversaturated passenger demand. Consequently, the performance score of scenarios S1 and S2 are significantly higher than those of scenarios S3 – S6. This demonstrates the importance of flexible fleet sizes in train routing and scheduling problems. Moreover, S1 outperformed the existing scheduling strategy (S7) with a 6.5% improvement in performance score, attributable to its flexibility of both routing and fleet sizes based on passenger demand in the transit network. These results indicate the potential of the model of S1 for application in real-world transit operations.

5.5. Sensitivity analysis

We first conduct a sensitivity analysis of the proposed model with respect to different demand variations under scenario S1. The demand rates across the entire network were adjusted to 80%, 90%, 100%, 110%, and 120% of their original values (see Fig. 7(b)). The key performance metrics derived from the computational results are presented in Fig. 14. Fig. 14(a) first shows that an increasing demand rate slightly raises the operation cost, but with a more significant increase in the passenger cost. Figs. 14(b) and



(a) Route 610 (S1-S4)

(b) Route 610 (S1 and S5-S7)

(c) Route 614 (S1-S4)

(d) Route 614 (S1 and S5-S7)

(e) Route 615 (S1-S4)

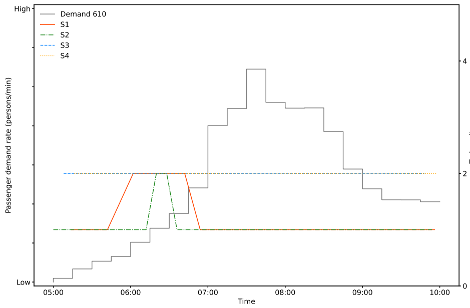
(f) Route 615 (S1 and S5-S7)

Fig. 12. Dispatching headways under different settings.

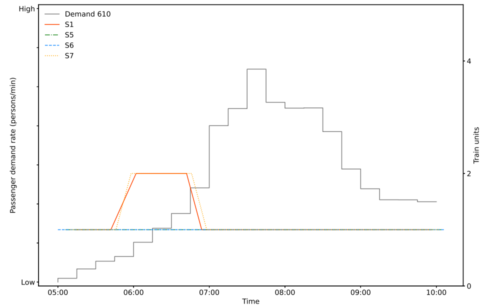
14(c) further reveal that both the total number of service runs and the average fleet size increased in response to higher passenger demand. Notably, when the demand rate decreased from 90% to 80%, the total number of service runs remained unchanged instead of decreasing. This can be explained by the model's effort to minimize passenger waiting times, which necessitated maintaining the total number of service runs. Meanwhile, the reduction in the average fleet size ensured that the operator's interests were under lower passenger demand.

We then conduct a sensitivity analysis of different numbers of driving modules and trailers. In this experiment, five different configurations of driving modules (= 38, 40, 42, 44, 46) and trailers (= 14, 15, 16, 17, 18) were adopted. Table 6 reveals the cost-saving advantage of deploying more train units (i.e., a larger number of driving modules and trailers). As the total number of train units increases, the model tends to dispatch more service runs while reducing the average fleet size, which results in a slight increase in operating cost but a significant reduction in passenger cost, particularly when the number of driving modules is fewer than 42. The overall results also reveal a diminishing marginal benefit from increasing the total number of train units. The cost-saving effects gradually decrease as the total number of train units continues to grow.

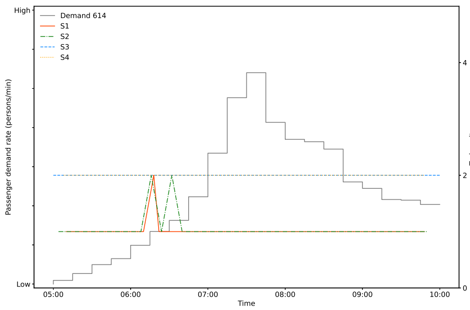
Additionally, we conduct a sensitivity analysis with respect to the operating cost coefficient ζ, which simulates slight variations in operating cost. In this study, five different values of ζ (= 0.90, 0.95, 1.00, 1.05, 1.10) were considered under scenario S1. Fig. 15



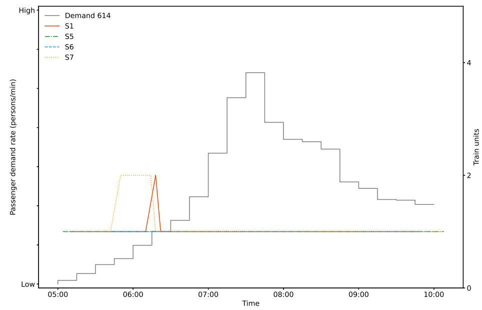
(a) Route 610 (S1-S4)



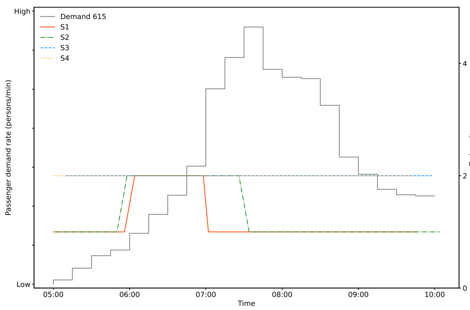
(b) Route 610 (S1 and S5-S7)



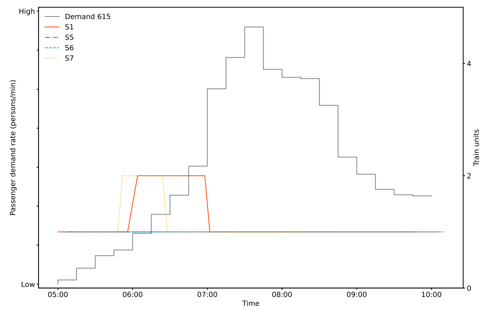
(c) Route 614 (S1-S4)



(d) Route 614 (S1 and S5-S7)



(e) Route 615 (S1-S4)



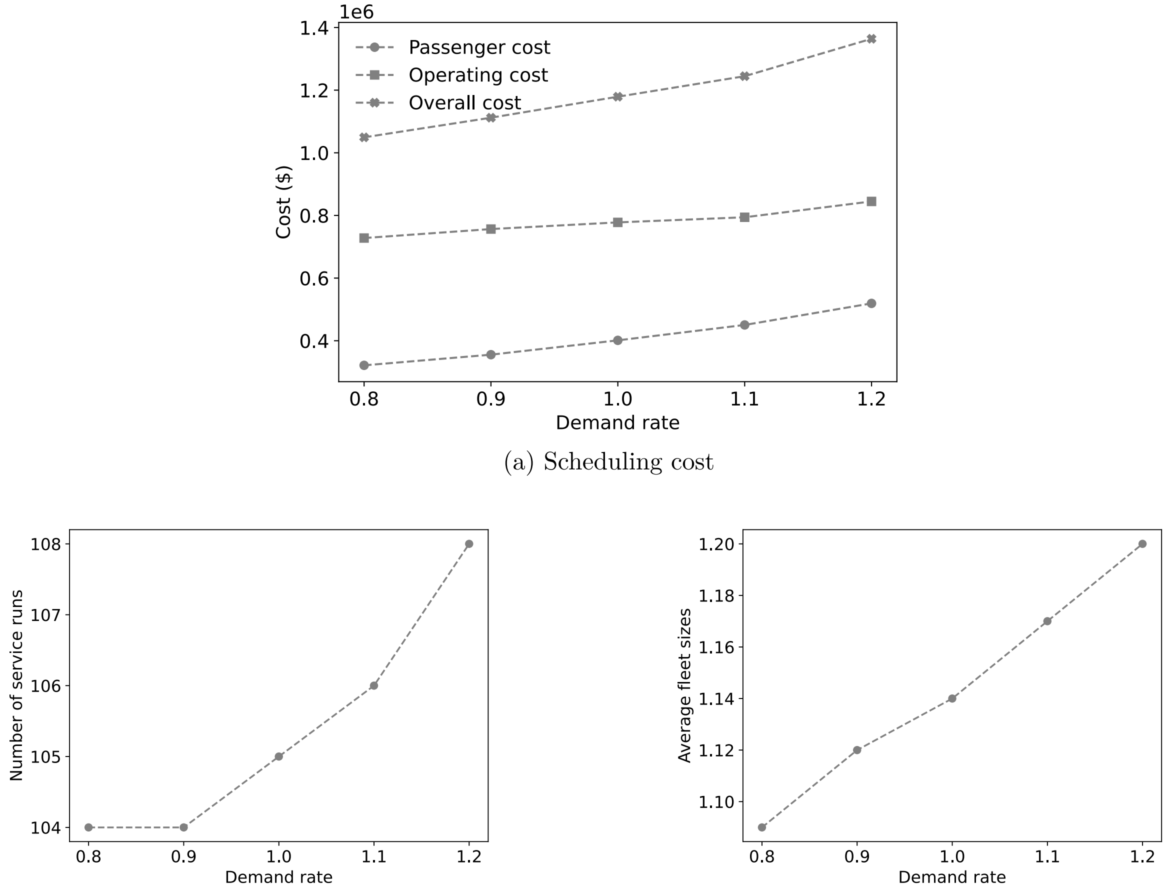
(f) Route 615 (S1 and S5-S7)

Fig. 13. Fleet sizes under different settings.

Table 6
Sensitivity analysis of different numbers of driving modules and trailers.

Driving modules	Trailers	Overall cost ($)	Passenger cost ($)	Operating cost ($)	Number of service runs	Average fleet sizes
38	14	1285254	497962	772292	99	1.19
40	15	1213274	438744	774530	102	1.17
42	16	1179056	401233	777822	105	1.14
44	17	1177866	396062	781804	107	1.13
46	18	1176759	391462	785297	108	1.13

presents the results of the sensitivity analysis, where the ratios of operating cost and passenger cost represent the relative changes in these costs compared to their values when $\zeta = 1$. The figure shows that both operating cost and passenger cost increase as ζ increases, but operating cost rises at a faster rate (i.e., with a steeper slope). This can be attributed to the direct impact of increased operating cost coefficients ζ. Moreover, the variations in passenger cost and operating cost demonstrate the proposed model's ability to balance



(a) Scheduling cost

(b) Number of service runs

(c) Average fleet sizes

Fig. 14. Sensitivity analysis of different demand rates.

these two factors automatically. These findings validate the proposed model's effectiveness in adapting to varying operating cost assumptions.

Finally, we conduct a sensitivity analysis to test the impact of the trade-off coefficient α, which balances the passenger cost and the operating cost in the reward function setting. This type of analysis is beneficial in multi-objective optimization as it reveals the sensitivity of different objectives to variations in the trade-off parameter (Chow et al., 2017; Chow and Pavlides, 2018). In this study, five different values of α (= 0.1, 0.3, 0.5, 0.7, 0.9) were considered under scenario S1. Fig. 16 presents the results of the Pareto analysis, with the passenger cost plotted against the operating cost. The figure shows that setting α within $[0.3, 0.5]$ leads to a satisfactory performance with minimized costs for passengers and operators. When the trade-off coefficient exceeds this range, such as when set at $\alpha = 0.1$ or $\alpha = 0.9$, the system gains limited benefits in one objective at the expense of the other. For this case study, the results suggest that setting $\alpha = 0.5$ is effective in reducing both passenger and operating costs.

6. Conclusion

This study presents an adaptive train scheduling framework with flexible fleet sizes for routing and scheduling in network-wide rail transit services. The objective is to minimize the total passenger waiting time and operating costs. The train scheduling problem is formulated as a partially observable Markov decision process (POMDP) to reflect the practicality in training and real-world applications. To address the computational challenges associated with the train scheduling problem, deep reinforcement learning techniques are applied to seek potential optimal solutions to the optimization problem. Specifically, an artificial neural network (ANN) surrogate is employed to approximate the value function, which estimates the expected future costs associated with the decisions made. The ANN surrogate enhances the computational efficiency of solving the optimization problem. The surrogate approximation is trained using a deep recurrent Q network (DRQN) training algorithm, which performs well in partially observable environments.

The proposed solution algorithm is tested using real-world scenarios and data collected from the Hong Kong Light Rail Transit network. The experiments first show that the DRQN-based training algorithm performs and converges well in training the ANN

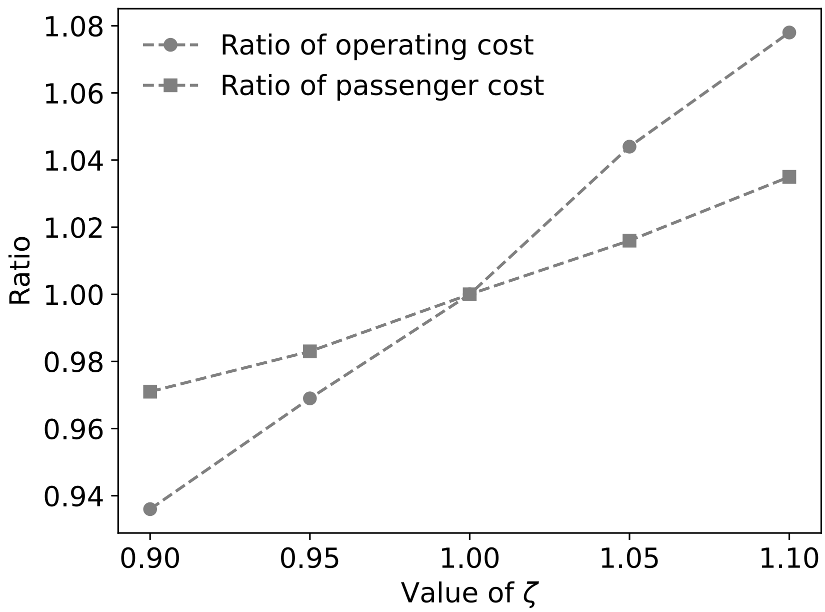


Fig. 15. Sensitivity analysis of different operating cost coefficients ζ.

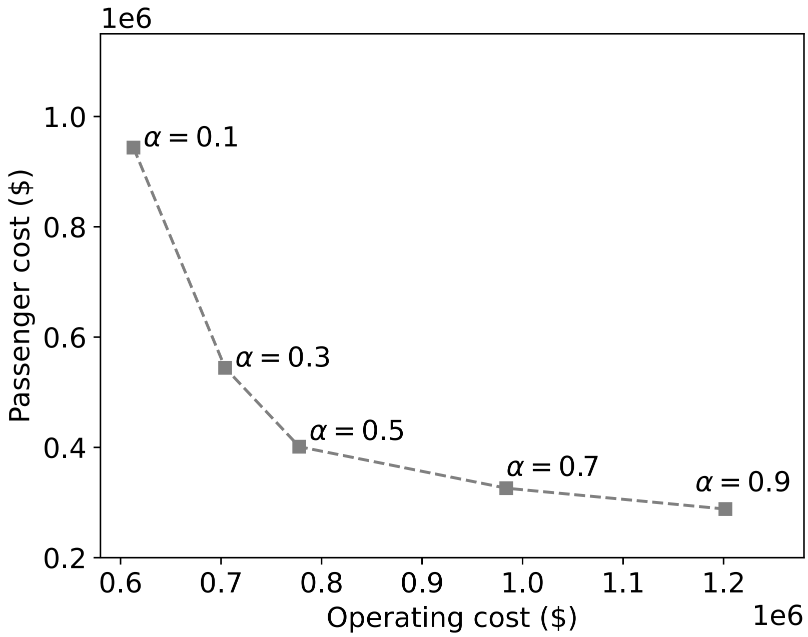


Fig. 16. Pareto solutions with respect to different trade-off coefficients α.

surrogate. By comparing with the well-established genetic algorithm (GA), particle swarm optimization (PSO), and deep Q network (DQN), the proposed reinforcement learning framework's effectiveness has been proven. Furthermore, the proposed train scheduling framework is compared with other experimental settings that offer less flexibility. The results demonstrate that the proposed train scheduling framework outperforms these settings in overall performance. That is mainly attributed to the adaptiveness of the proposed model, which allows for dispatching small fleet sizes (single-car) during off-peak periods to reduce costs and employing a flexible routing strategy during peak periods to minimize passenger waiting times. These findings suggest the importance of having flexible fleet sizes and routing for optimizing the operations of a transit network throughout different times of the day.

Reinforcement learning has demonstrated significant potential in developing efficient urban transportation systems. Several prospective research directions have been identified. Firstly, we aim to extend the train scheduling framework to general road-based transportation services, such as bus services, with adaptive routing and ride-sharing. Moreover, the train scheduling framework presented in this study adopted a centralized approach, where a single central controller makes decisions. A future research direction involves expanding this centralized framework into a decentralized framework, enabling multiple controllers to derive and implement decisions at multiple depots (Ying et al., 2022), and applying it to more complex rail transit networks. The challenge of a decentralized framework is ensuring overall effectiveness and stability from both training and implementation perspectives. Further developments in decentralized frameworks will be reported in the future.

CRediT authorship contribution statement

Shou-yi Wang: Writing – original draft, Writing – review & editing, Visualization, Software, Investigation, Validation, Methodology, Conceptualization. **Andy H.F. Chow:** Writing – review & editing, Supervision, Investigation, Conceptualization, Writing – original draft, Methodology, Funding acquisition. **Cheng-shuo Ying:** Writing – review & editing, Methodology, Funding acquisition, Writing – original draft, Investigation, Conceptualization.

Acknowledgments

The authors would like to thank the anonymous reviewers for the constructive feedback. The first and second authors are supported by the General Research Funds (GRFs: 11203822; 11217824) sponsored by the Hong Kong Research Grants Council (HKRGC). The third author is supported by the National Natural Science Foundation of China (72301274) and China Postdoctoral Science Foundation, China (2023M733688).

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